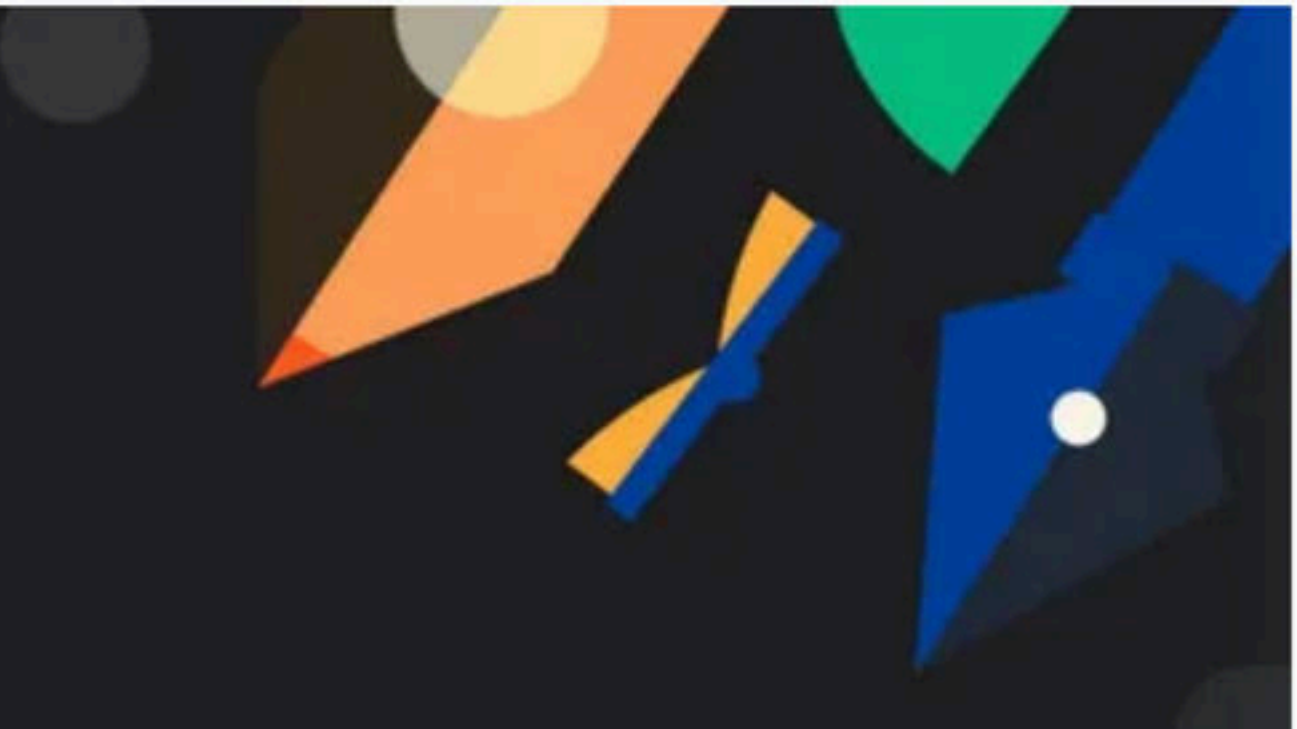




Doubt Class with Lakshay

Special class

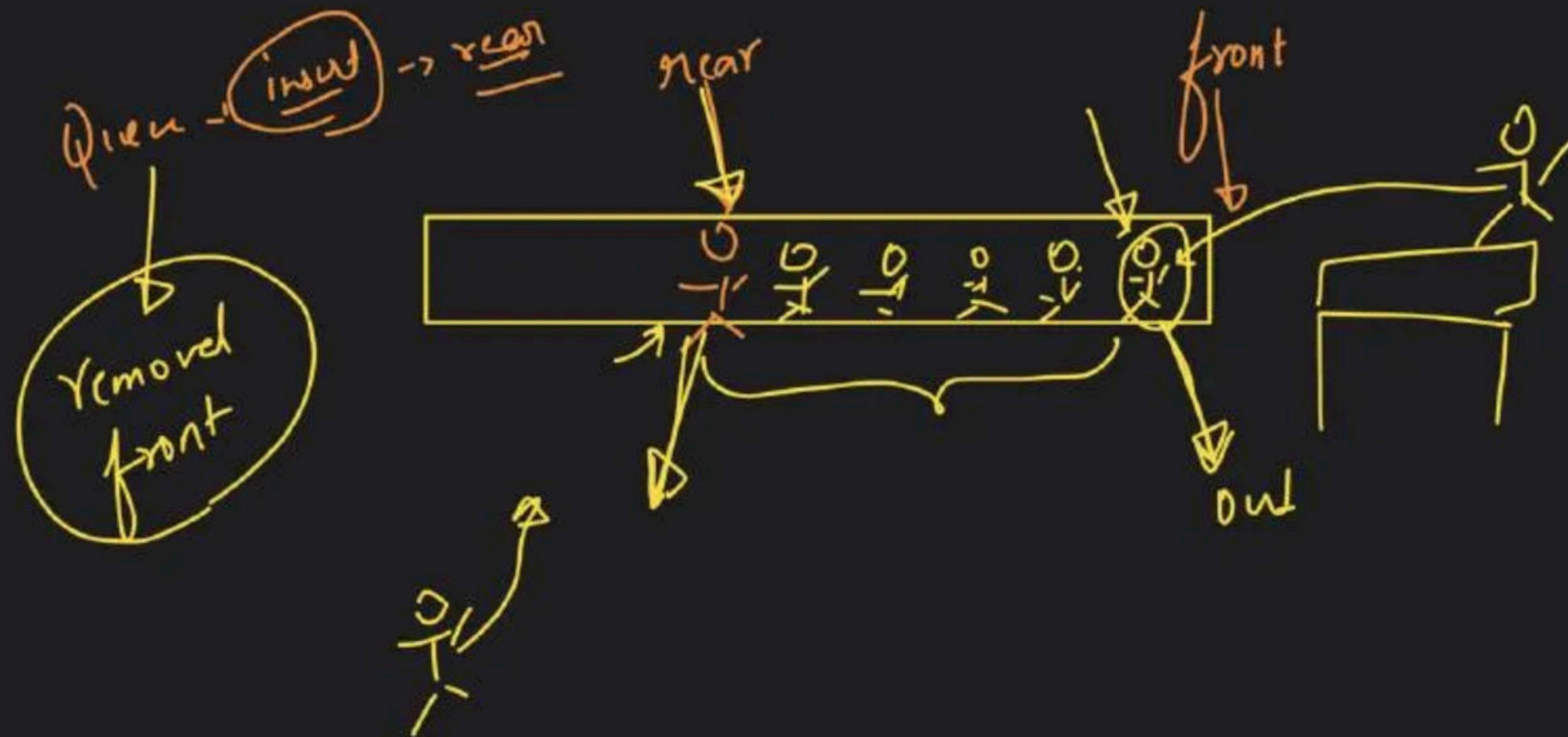
Stack Mega class
11 Aug 2024

Queue Class-1

Special class

=> Queue Doubts

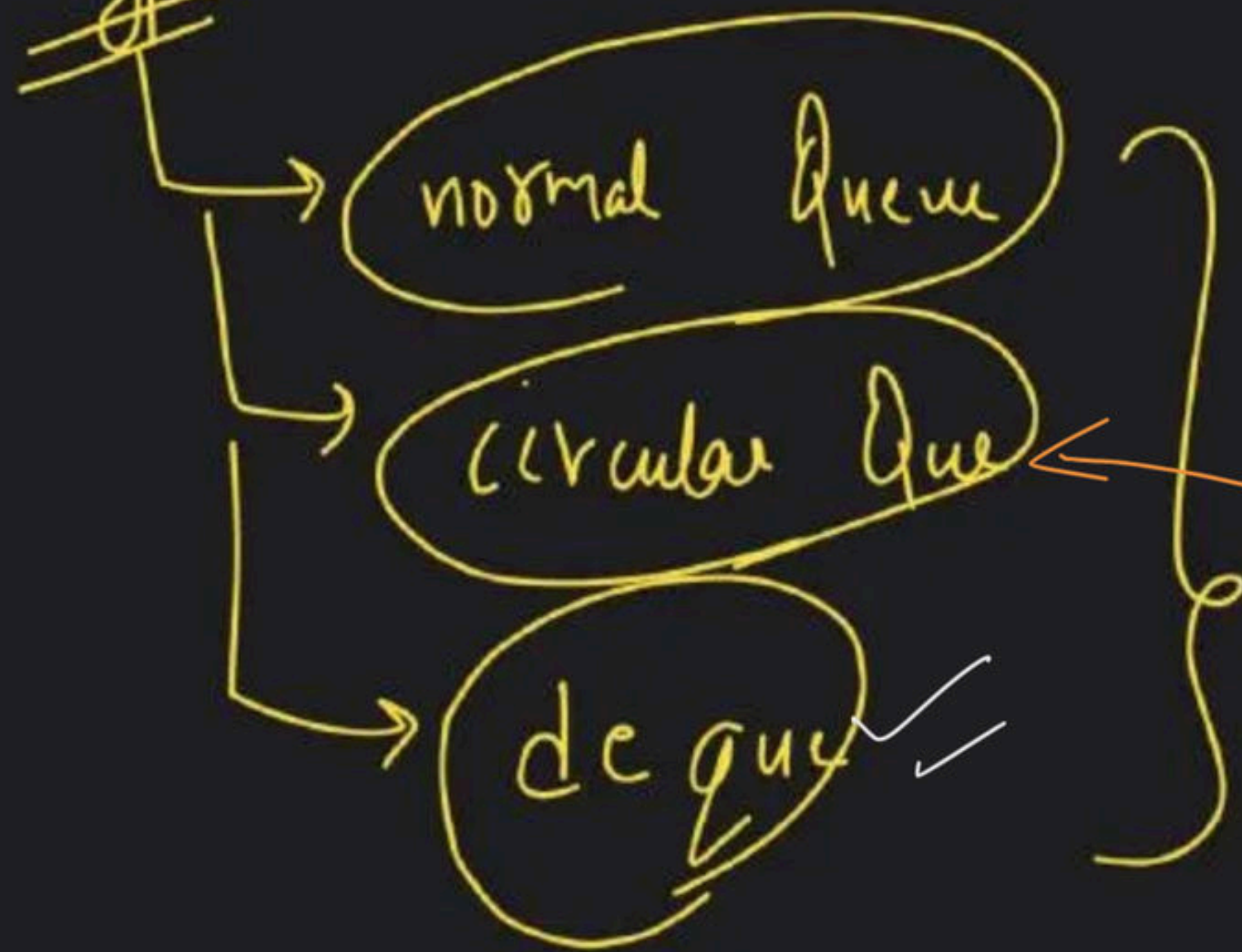
→ Queue → DS → FIFO



~~Types~~ inefficient ← Not used X



STL



↓ ✓
Queue < >

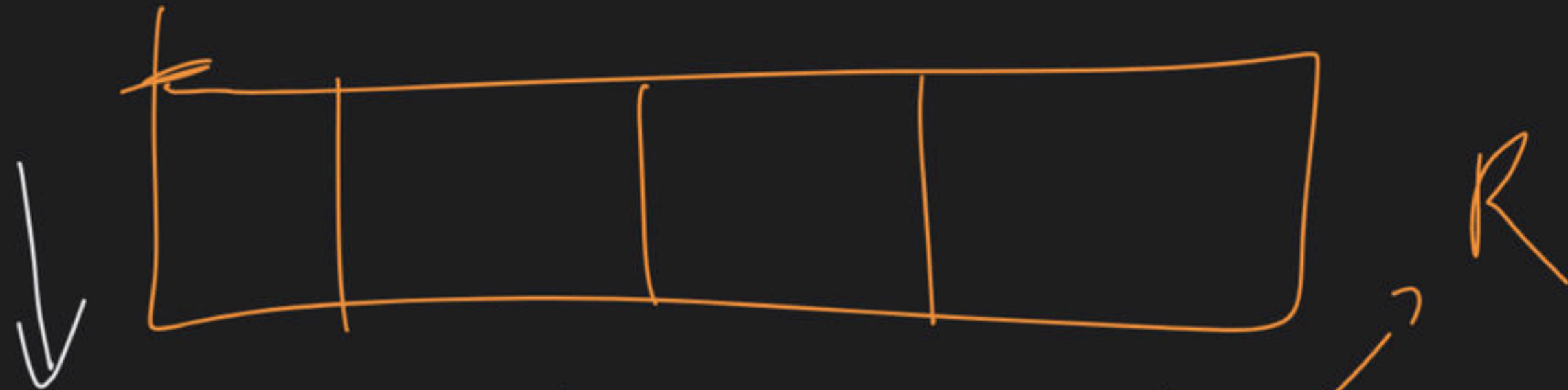
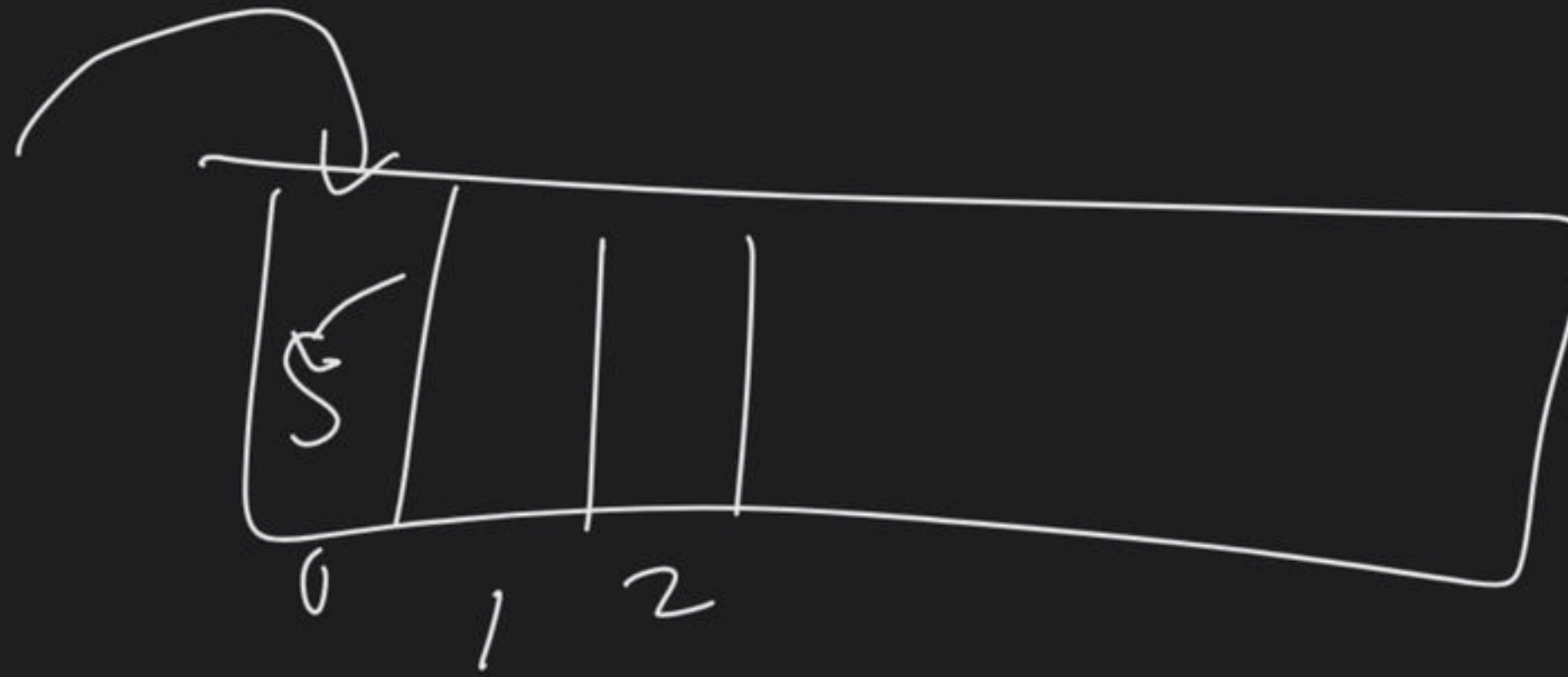
↙ ↘
Circular

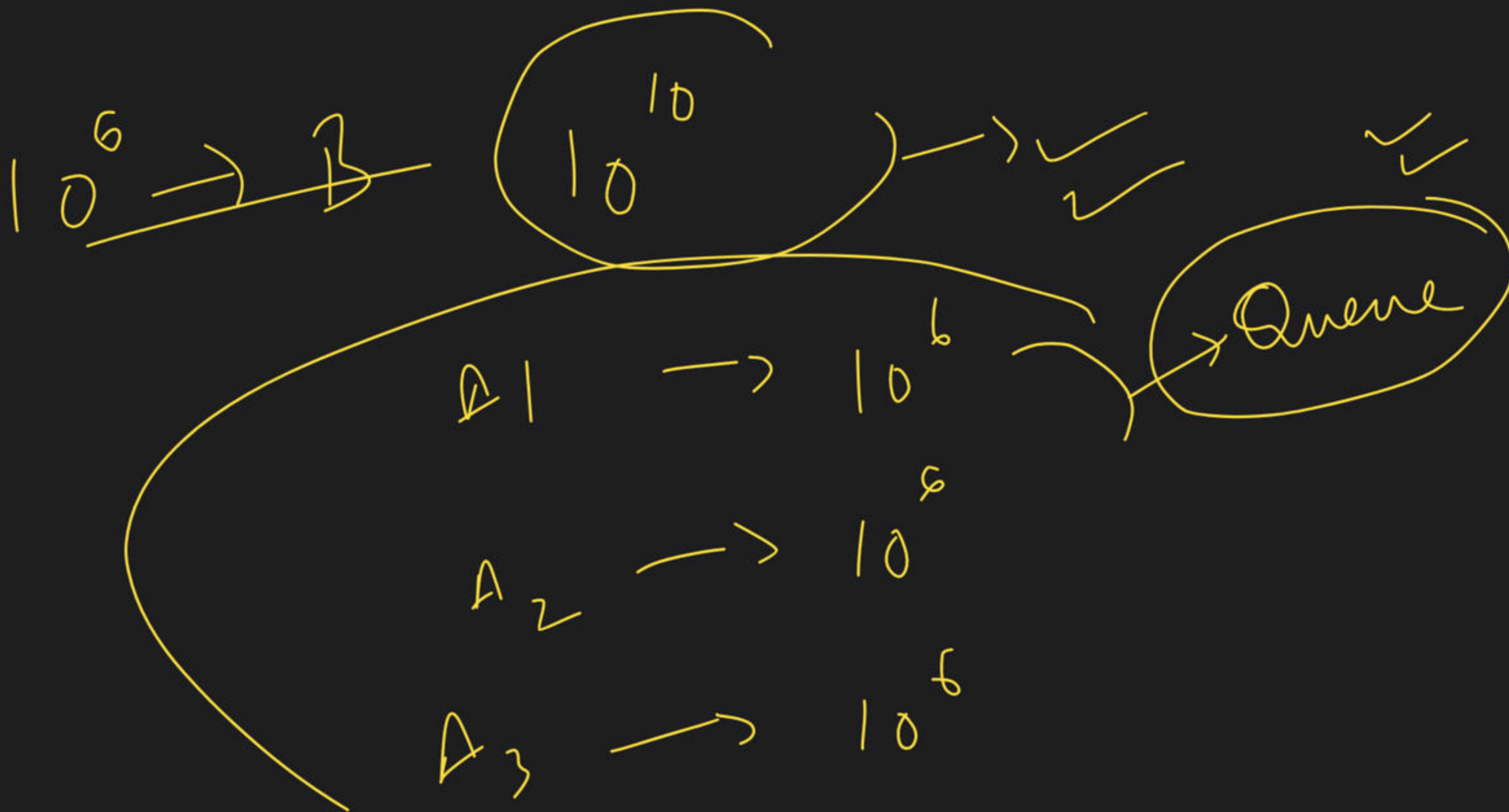
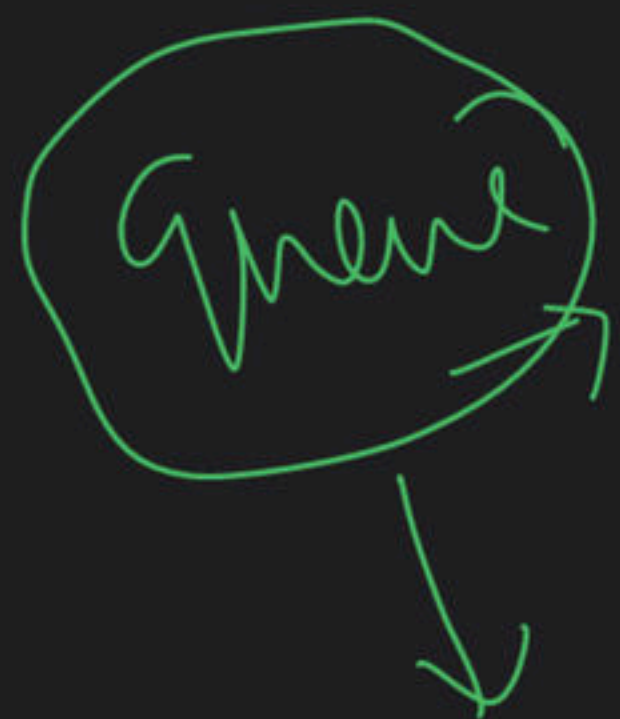
array



$$f = k - 1$$

$$R = \dots$$

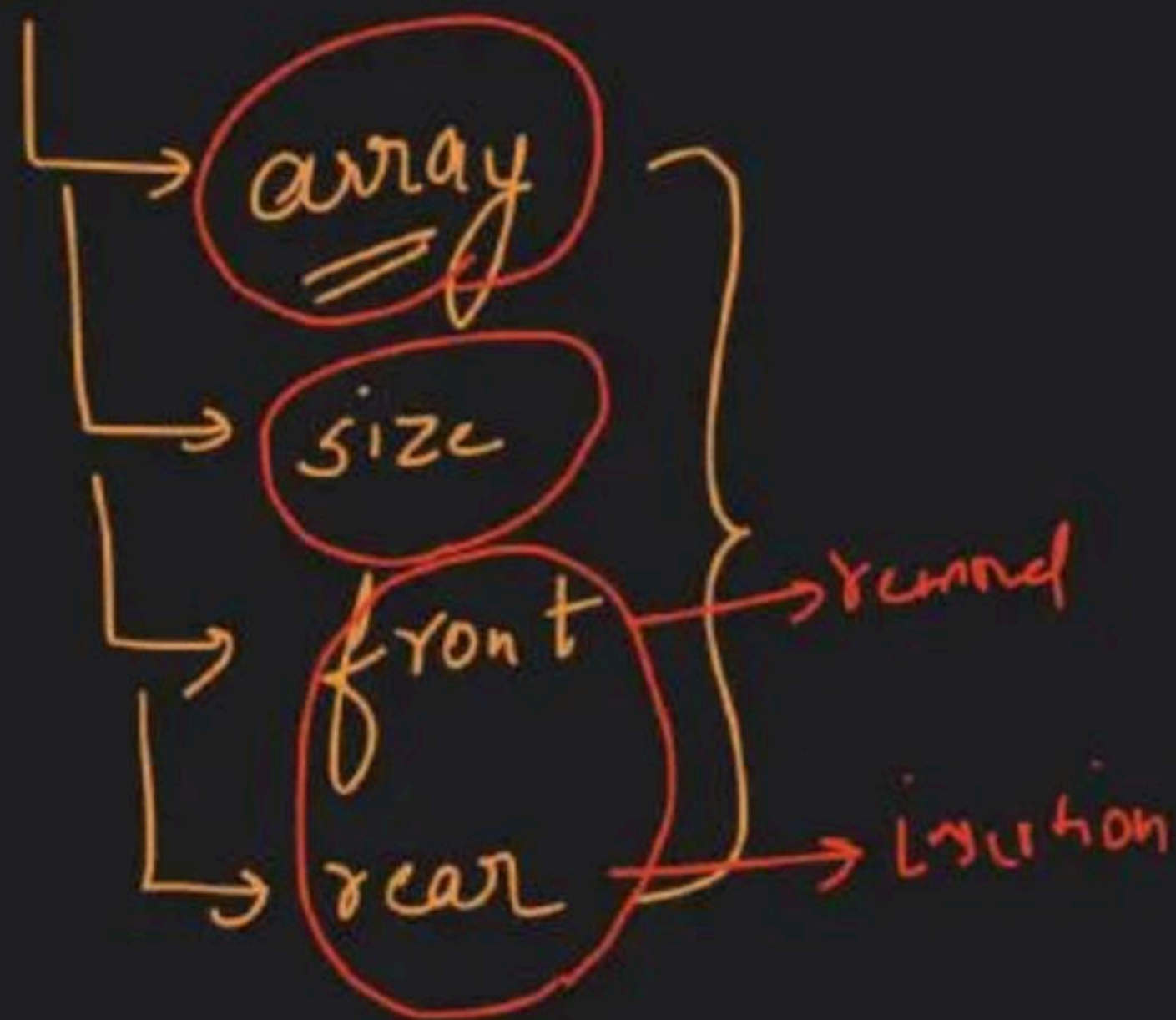


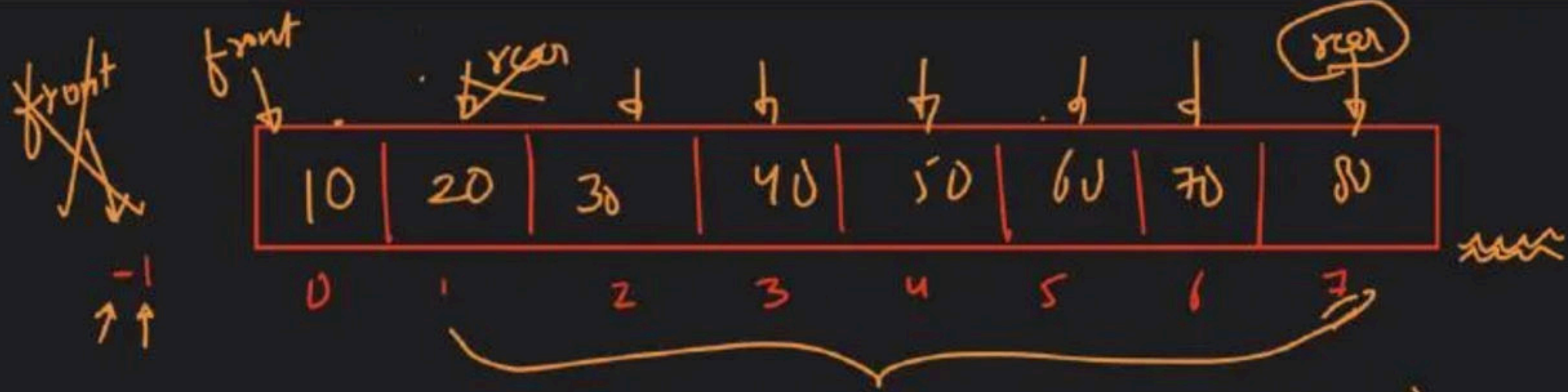


① stl
↓

(STL)

→
Scratch





insertion → rear → ① Overflow → If $(\text{rear} = \text{size} - 1)$ and $\text{count} < \text{size}$ Overflow

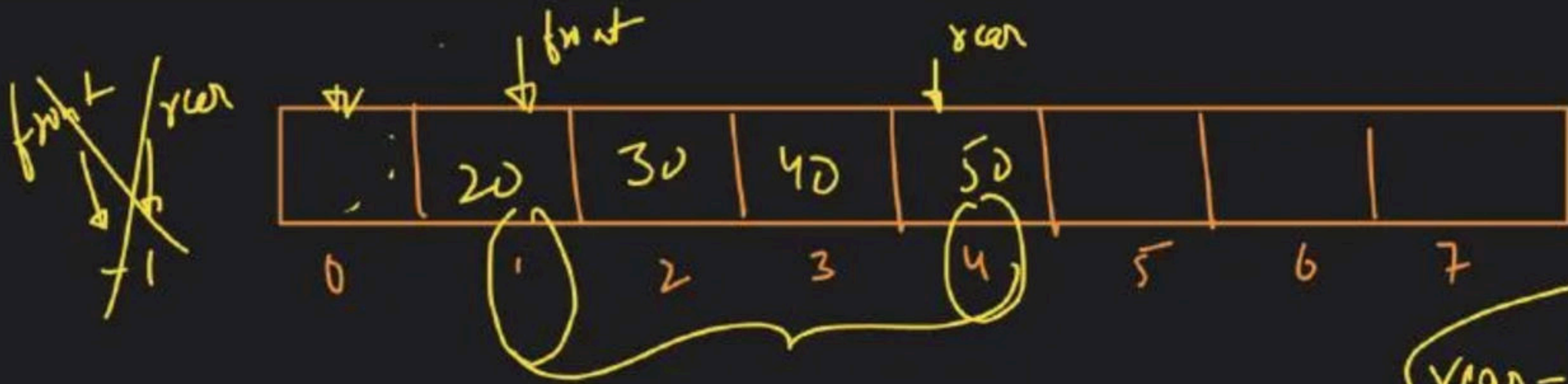
② First element → $\text{rear}++$, $\text{arr}[\text{rear}] = 10$, $\text{front}++$

③ normal insertion

2. push(10)
↑
2. push(20)

2. pop(100)

→ $\text{rear}++$
→ $\text{arr}[\text{rear}] = \text{val}$



$$\text{rear} - \text{front} + 1$$

$$4 - 1 = 3 + 1$$

$$= 4$$

pop

Empty → Underflow

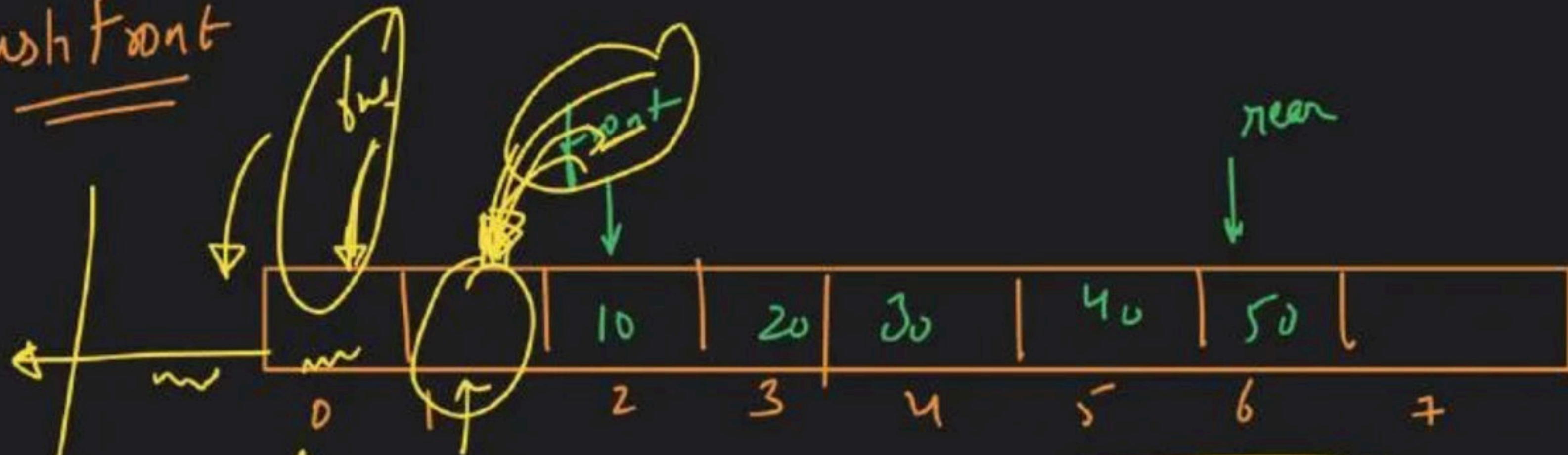
Single element

if (front == rear)

normal flow → arr[front] = -1
front++

2 min
+ 3 wall

→ push Front



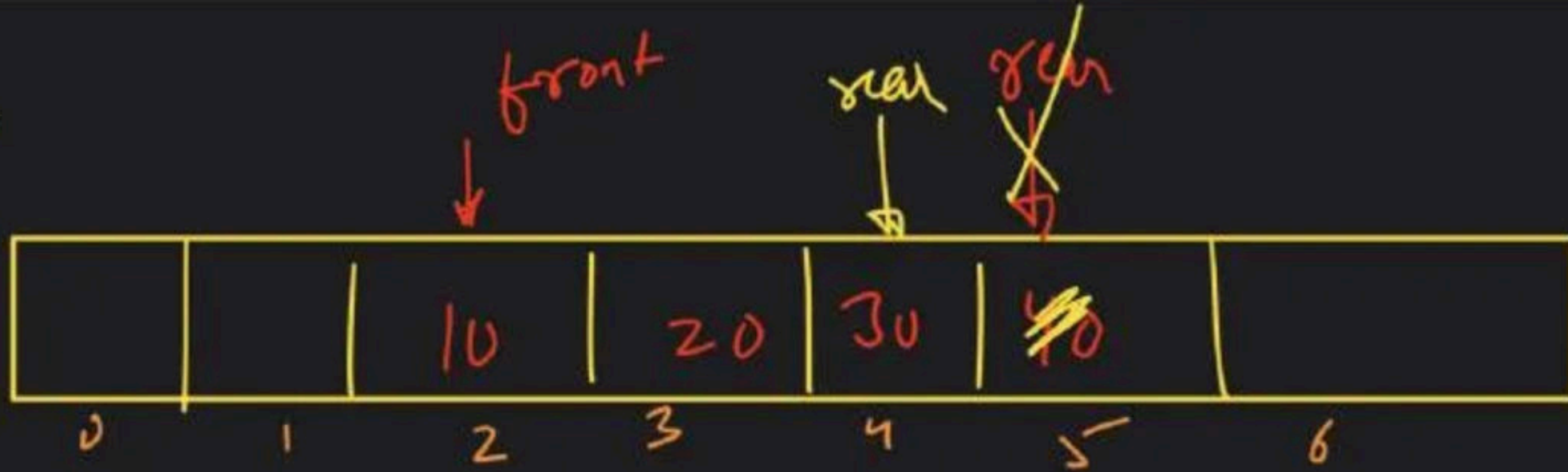
Overflow → if (front == 0) → Overflow

front -> if (front == -1 & rear == -1)

→ $f++$, $r++$, $arr[f] = val$

normal flow → f--, $arr[f] = val$

→ Popback

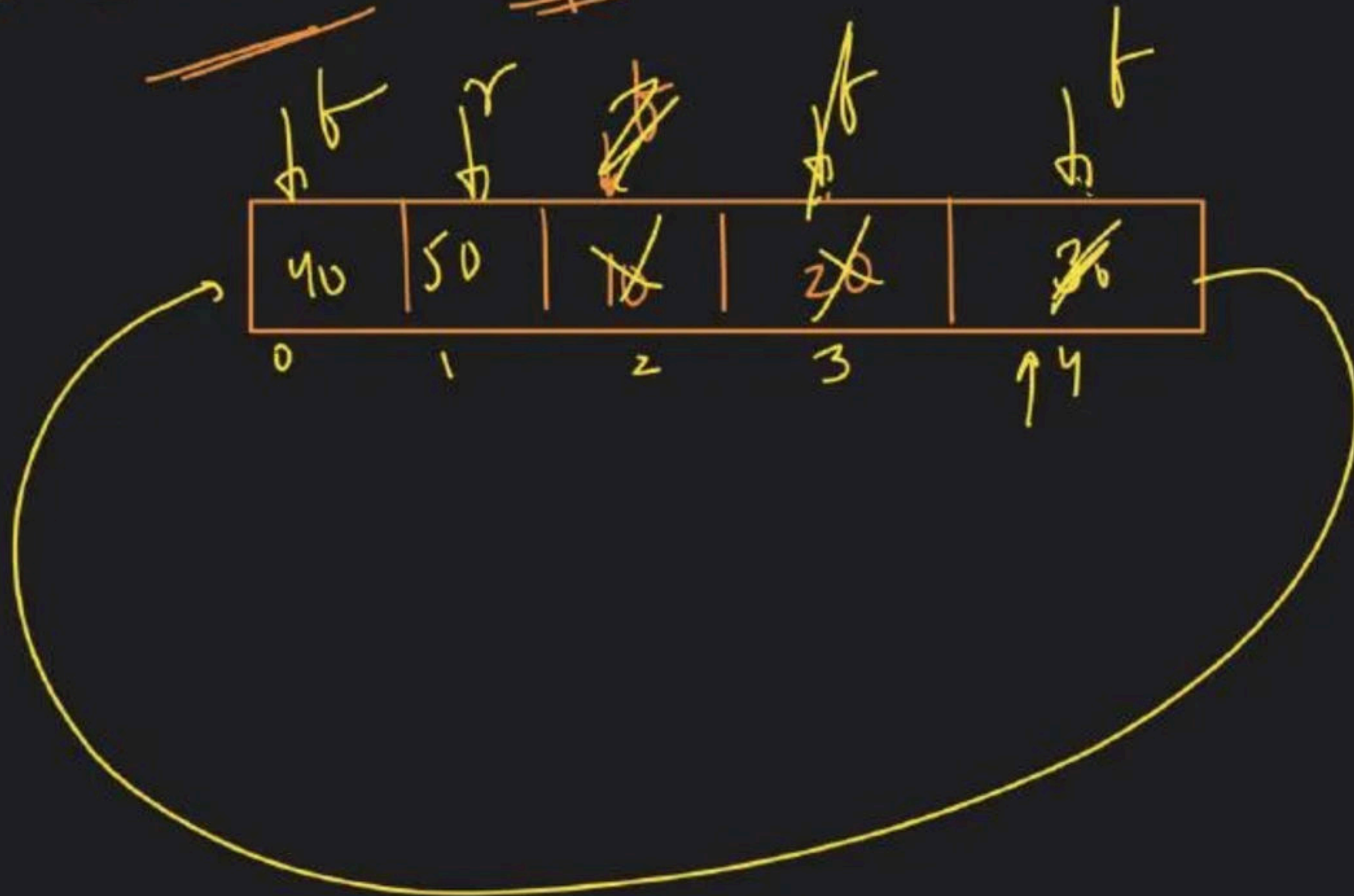


→ Underflow → $f = -1 \wedge r = -1$ → Underflow

→ $\text{is_empty} \rightarrow \text{if } (f == r) \rightarrow f = -1, r = -1$

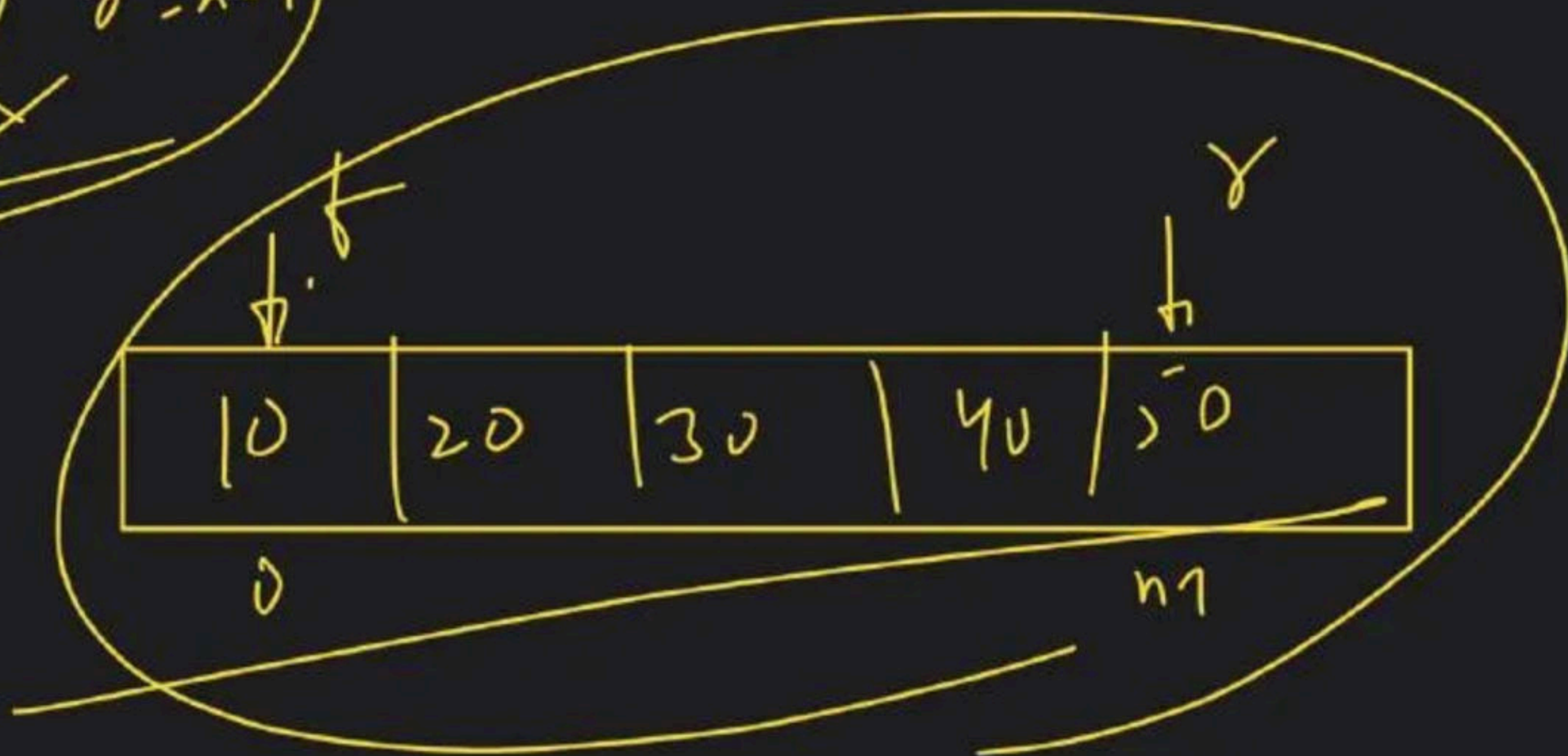
↳ now flow → $\text{rear}--$

→ Circular Queue

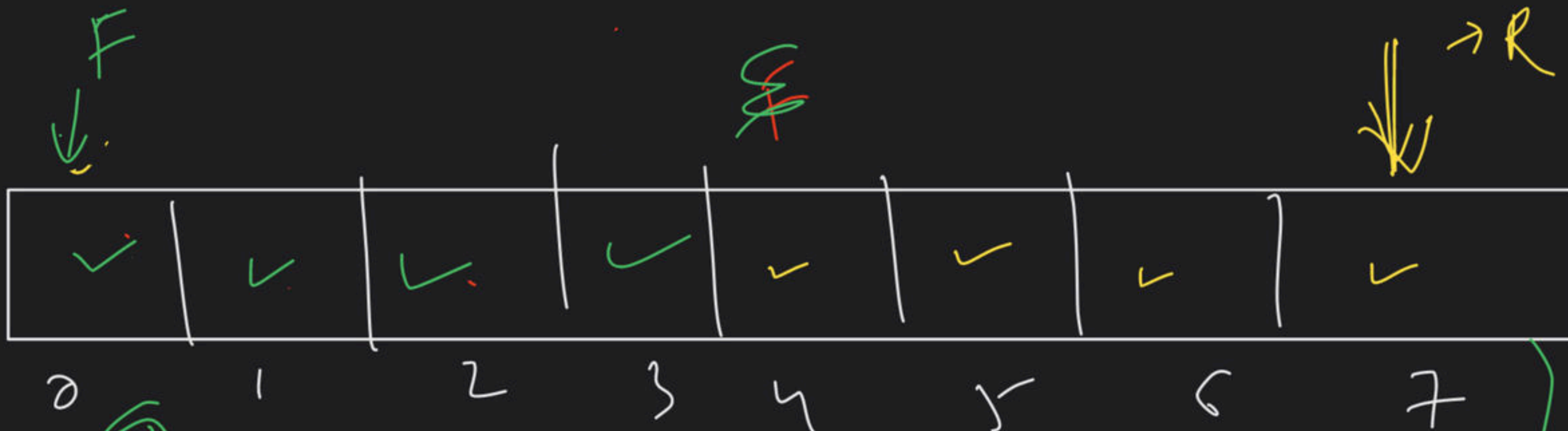


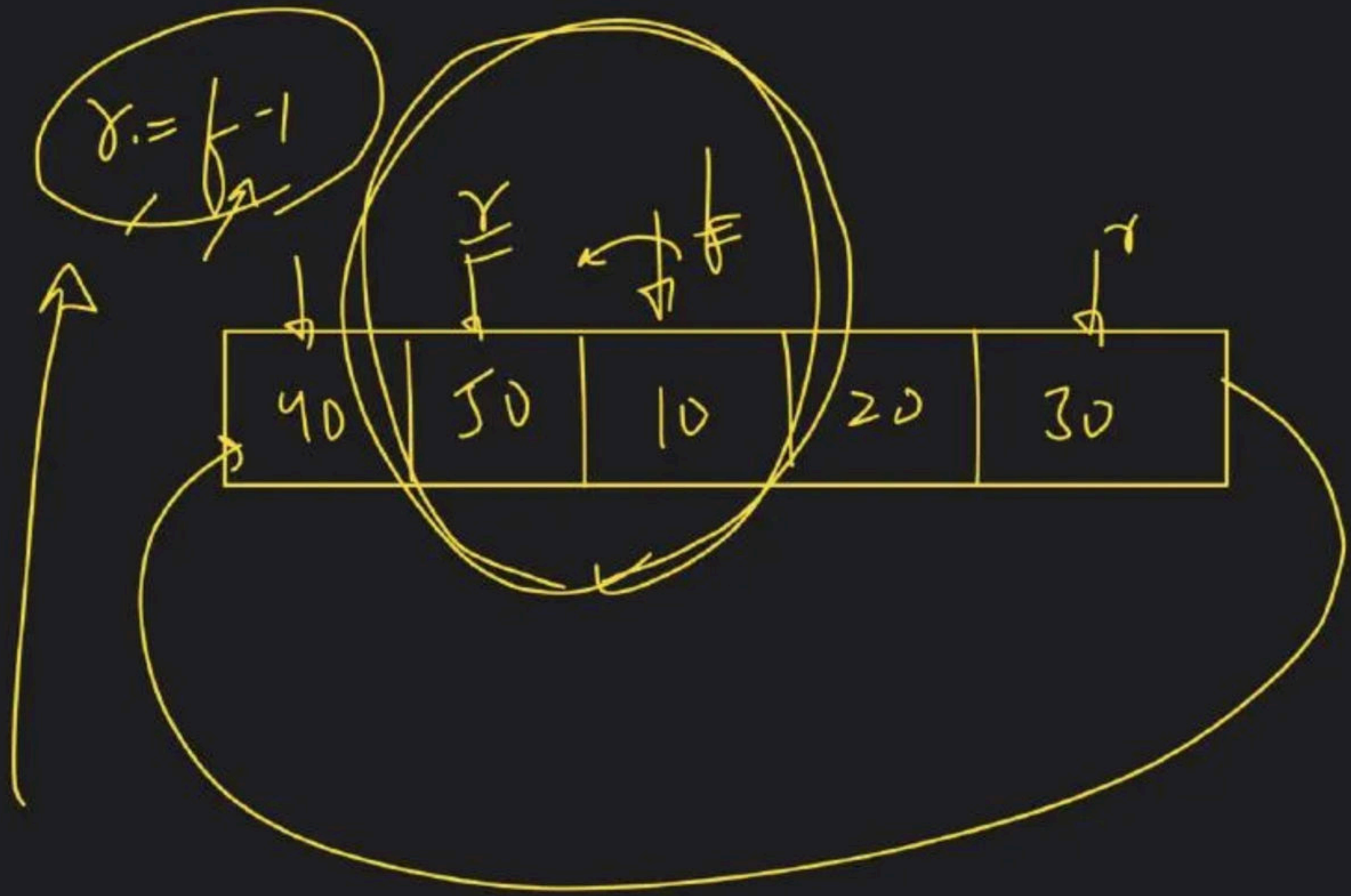


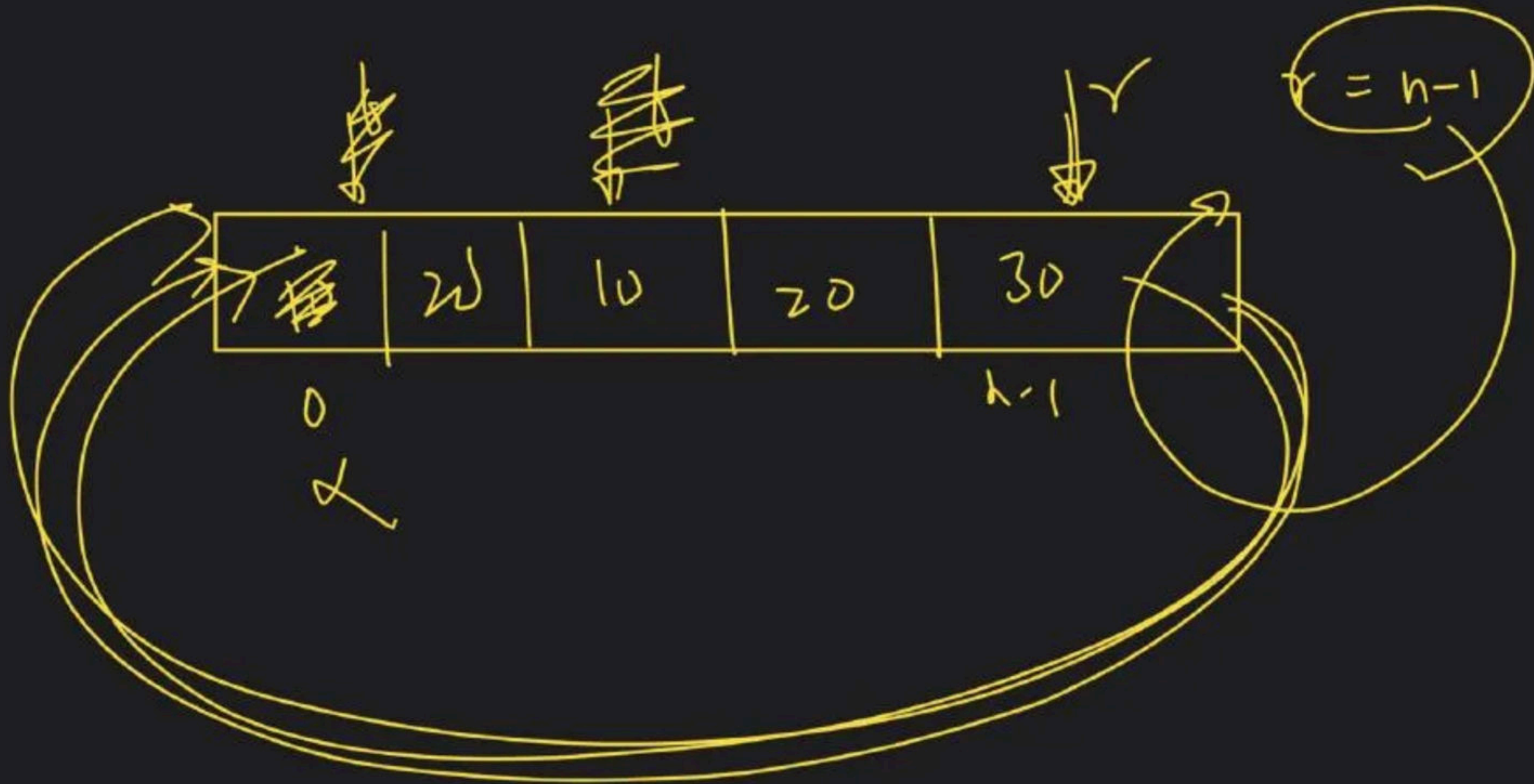
$l = 0$ & $r = n - 1$



$$f = -1$$
$$R = -1$$







pop

Underflow →

$f = -1$ & $r = -1$

Single elem →

$f = r$

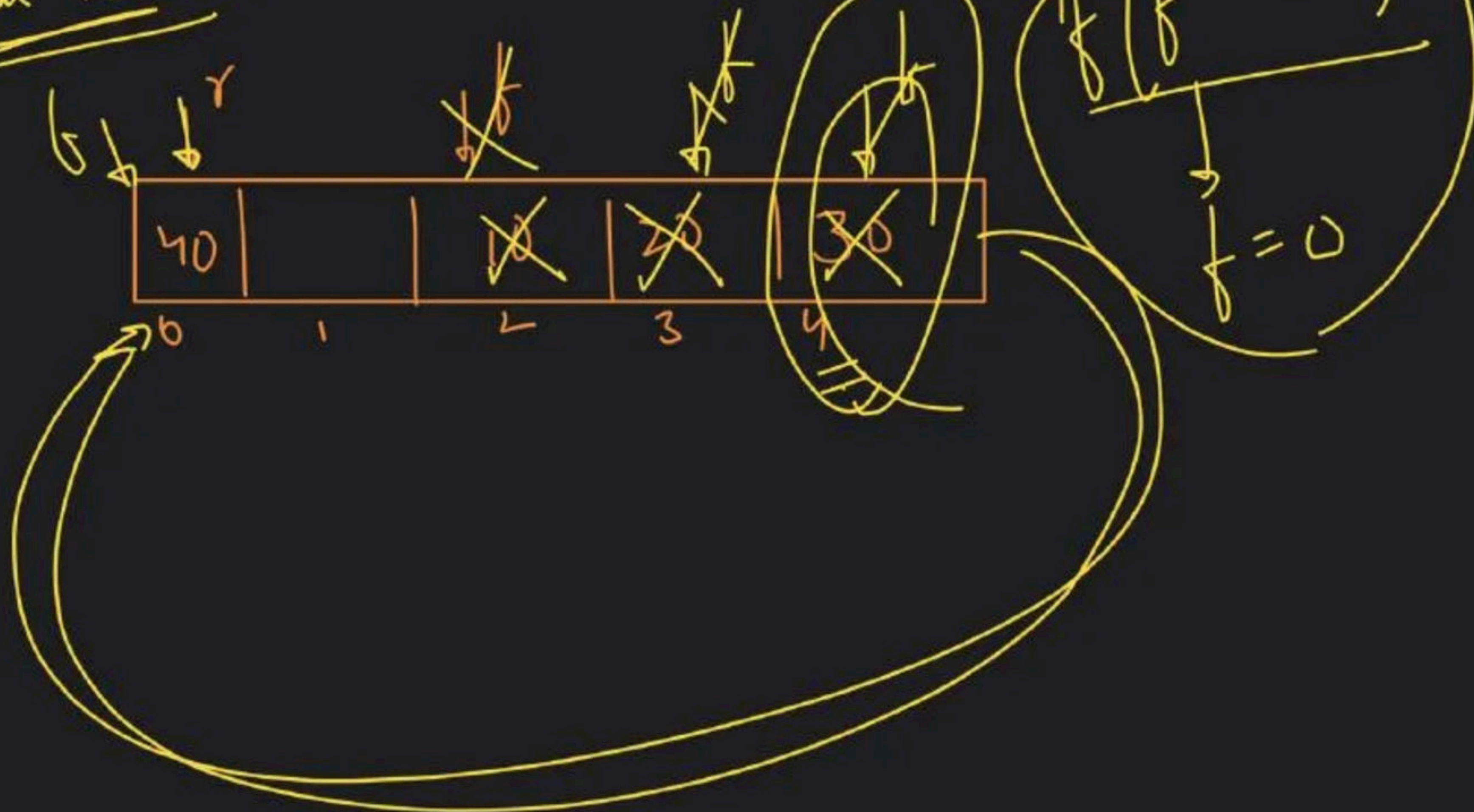
CiV

nl. F →

!
front++

$f = -1$ & $r =$

circular nature



			10	20	30
--	--	--	----	----	----

0 1 2 3 4 5

$$r - f + 1$$

$$5 - 3 + 1 = 2 + 1 = 3$$

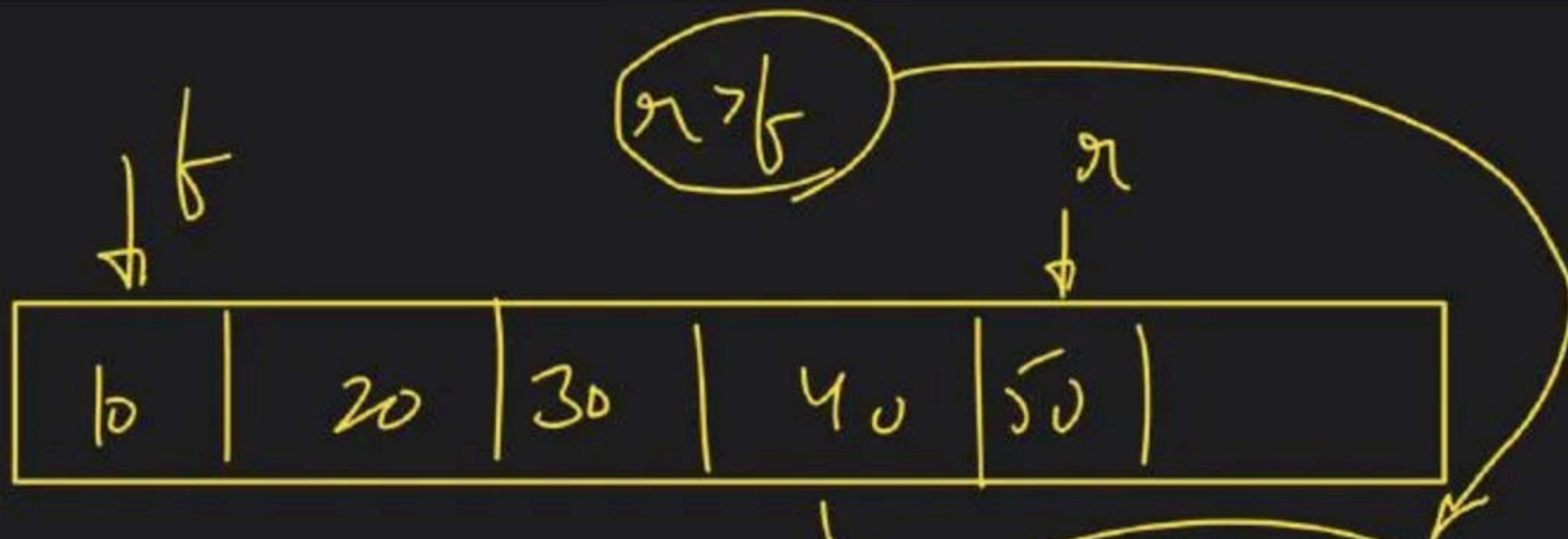
$$\text{abs}(r - f) + 1$$

40	50	30	20	10	40
----	----	----	----	----	----

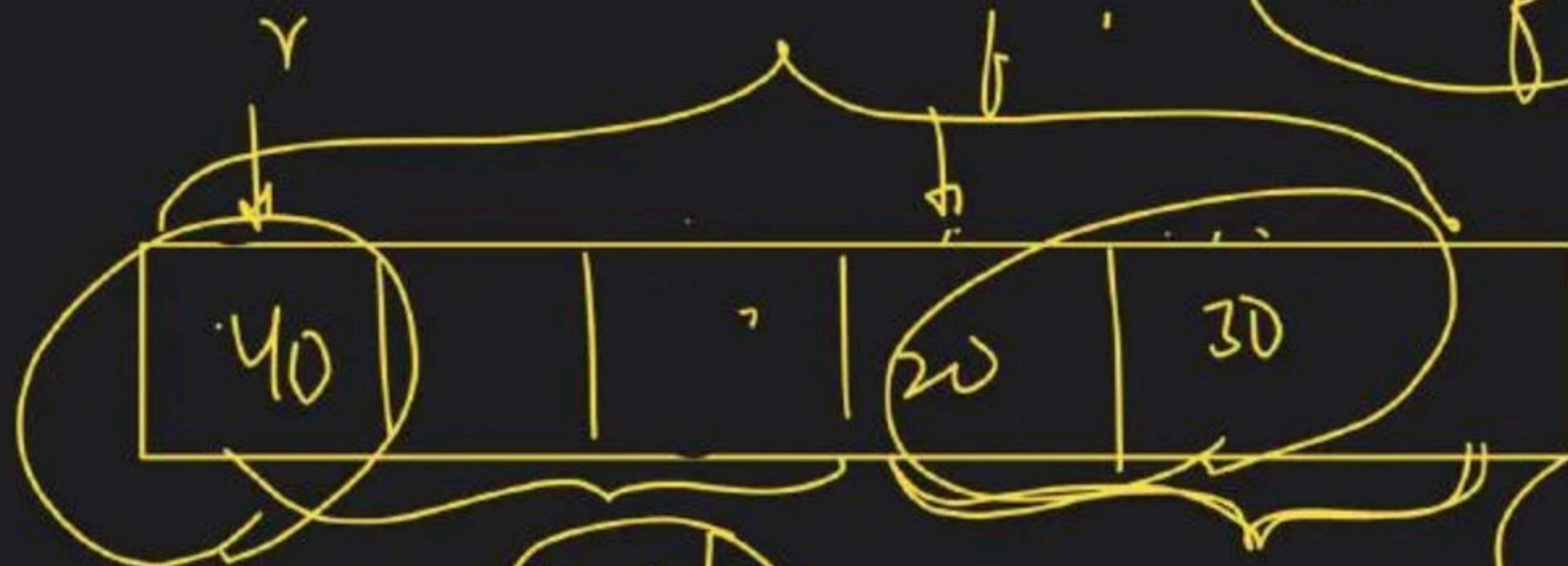
0 1 2 3 4

$$f - r + 1$$

size



$r - f + 1$



$size - f = 2$
 $5 - 3 = 2$

$r - f + r + 1$

$r+1$

		r		b						
10		20		30		40		50		6
0		1		2		3		4		5

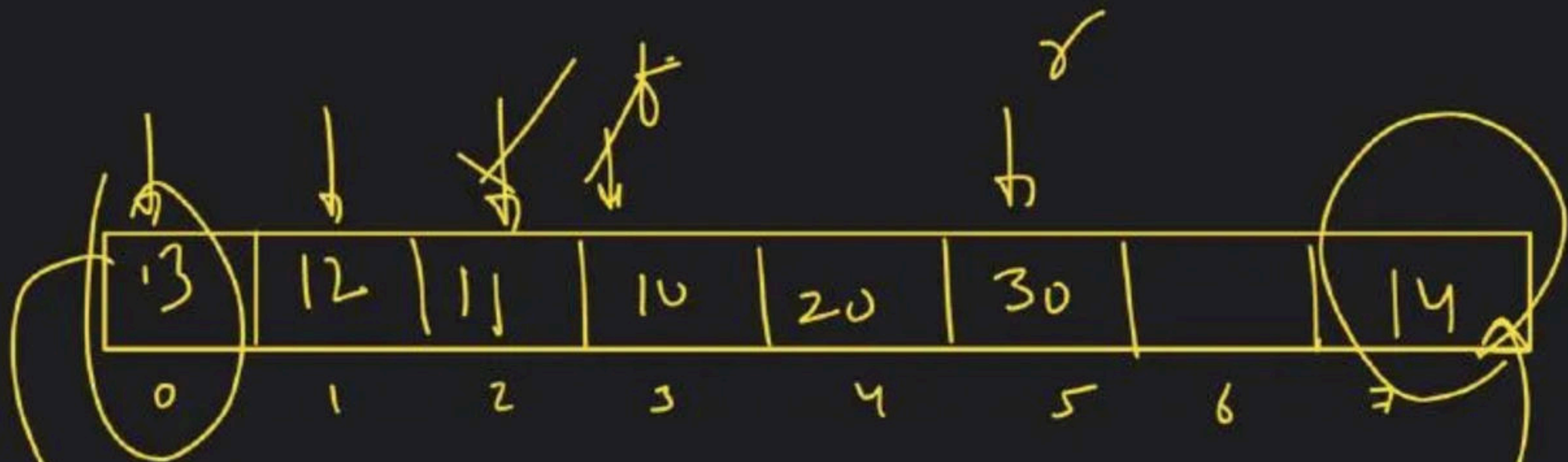
$$s = \frac{1}{b} + r + 1$$

$$6 = 3 + 2 + 1$$

$$= 6$$

degree + circular Nature

9 min
Break

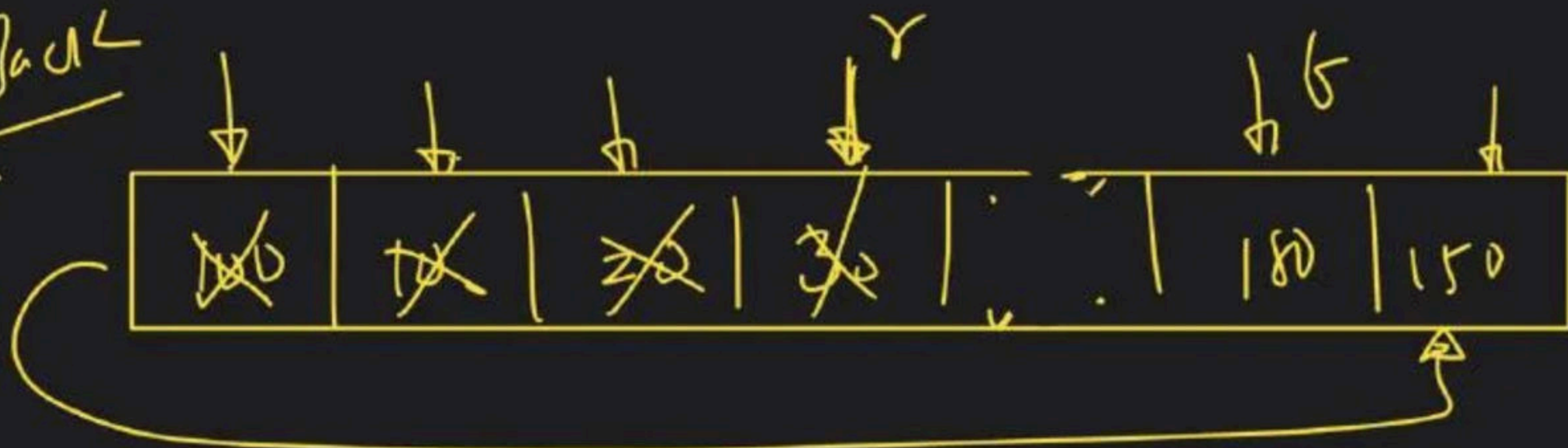


push front

normal $\rightarrow f \dots arr[f] = val$

$\rightarrow$ Circular $\rightarrow f = 0 \text{ \& \& } r = n-1$
 $\downarrow$
 $f = n-1$
 $arr[f] = val$

popBack



normal $\rightarrow$

$arr[r] = -1$, $r--$

Circular case $\rightarrow$

if $(rear == 0)$

$arr[r] = -1$
 $r \rightarrow n-1$

Queue Class - 2

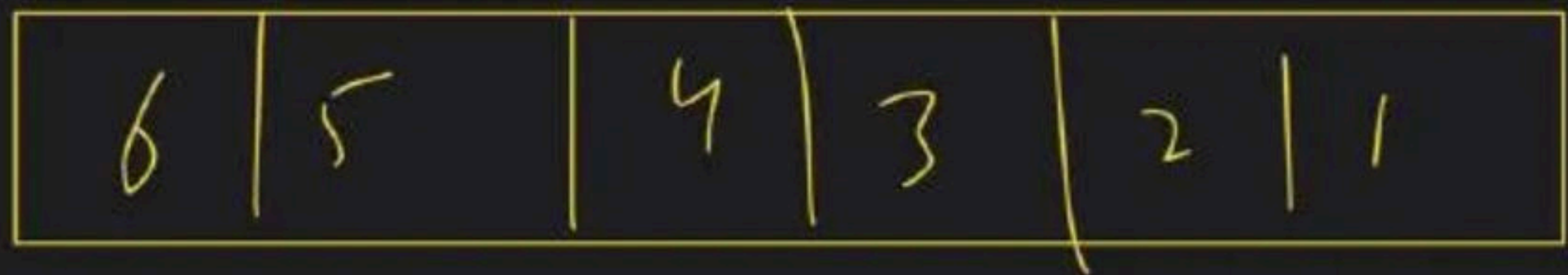
Special class

→ Reverse a Queue

STL

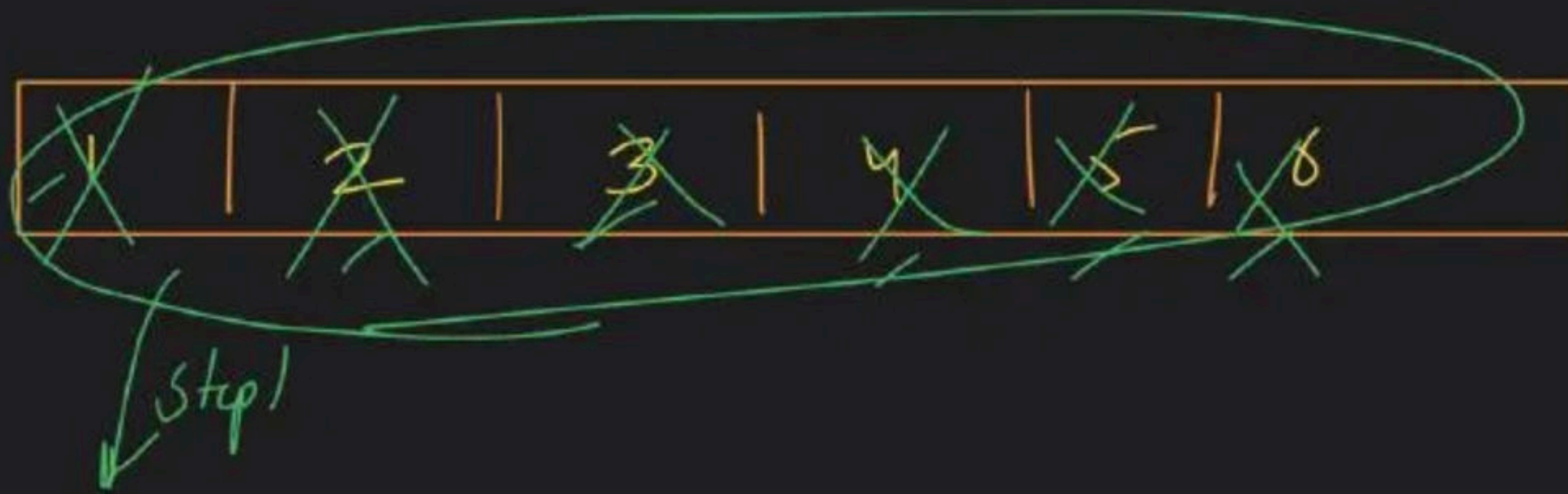
→ Rec //

→ Stack //

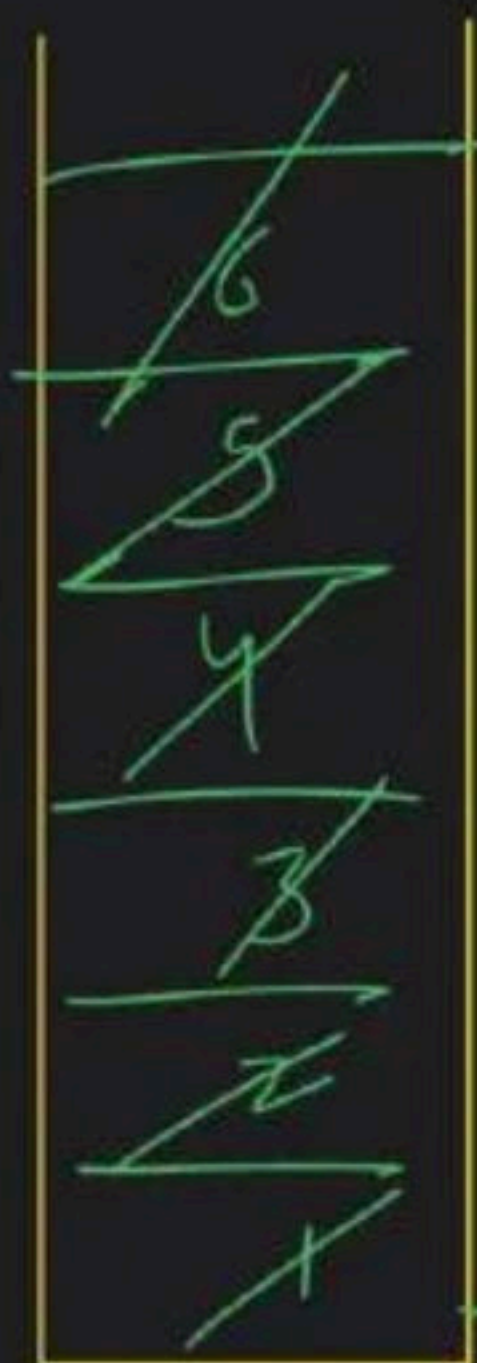


o/p

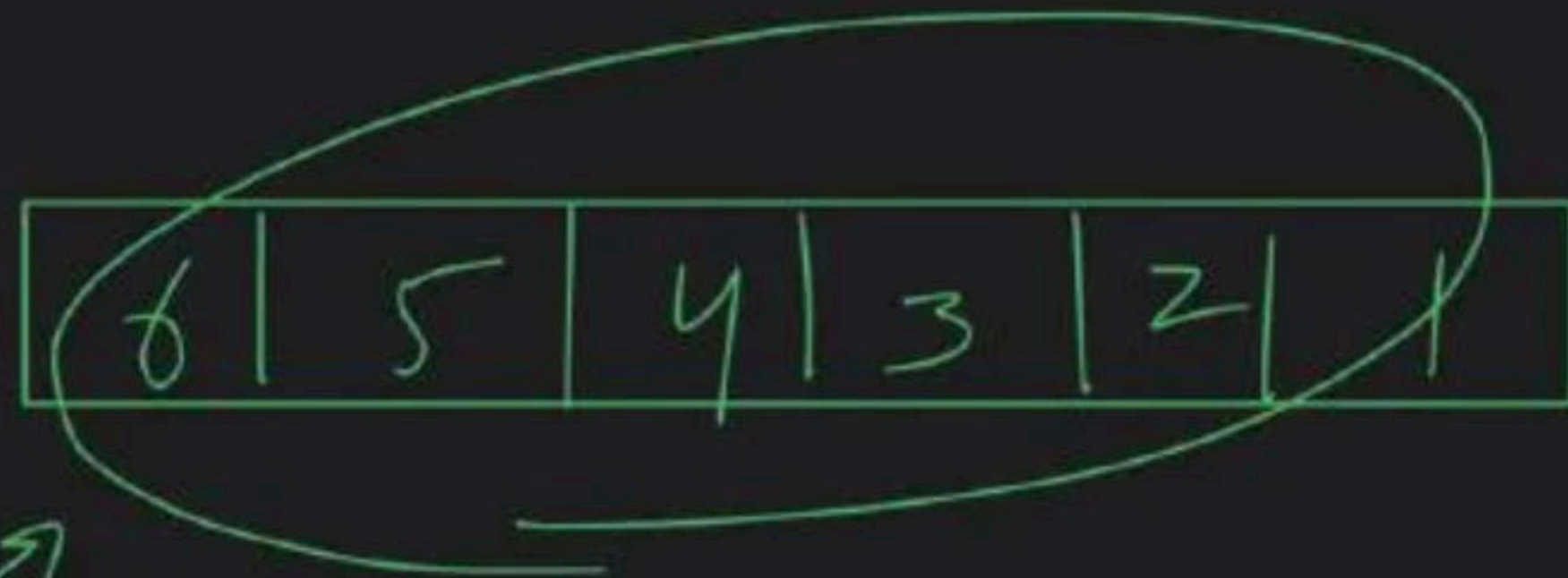
2

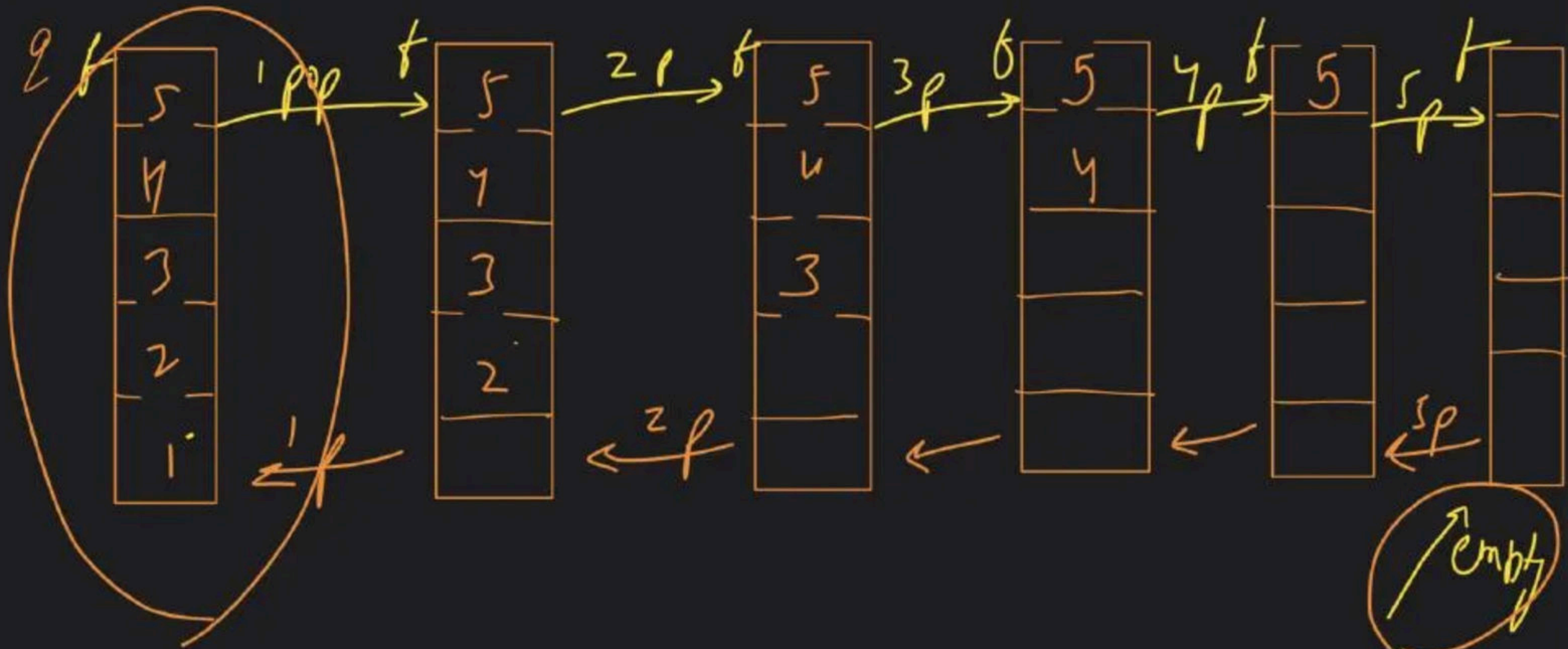


Stack



2

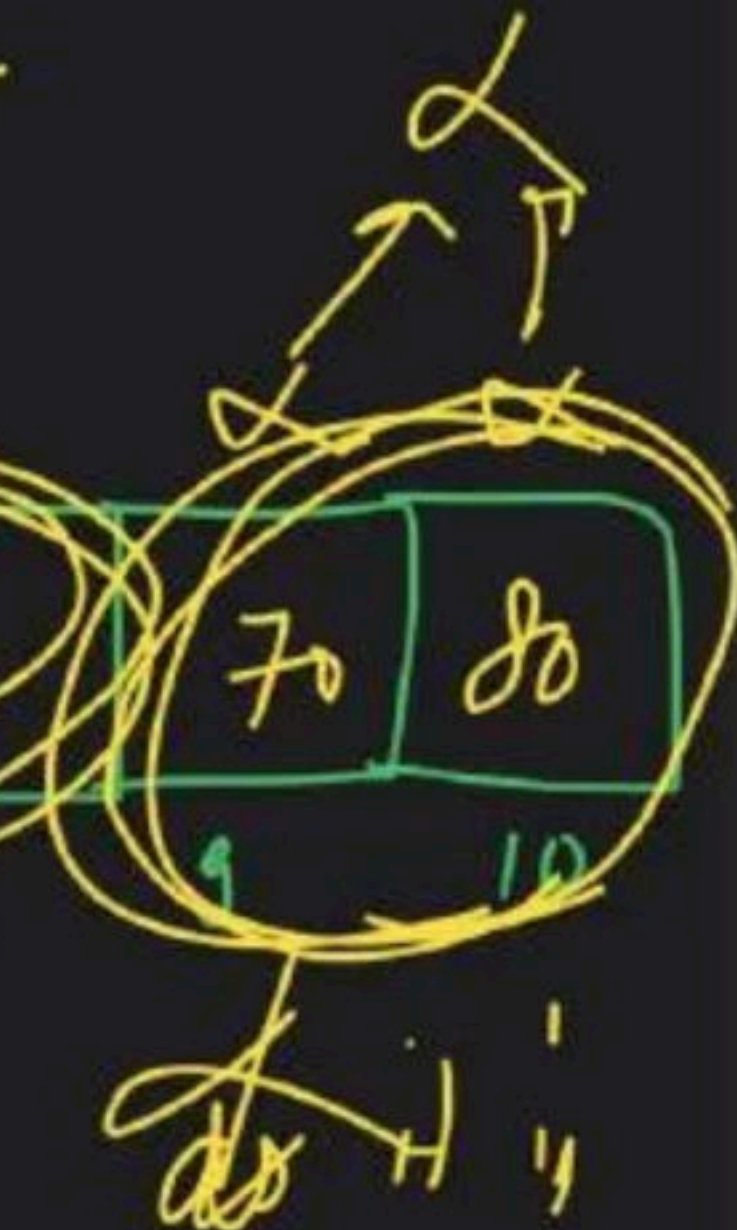
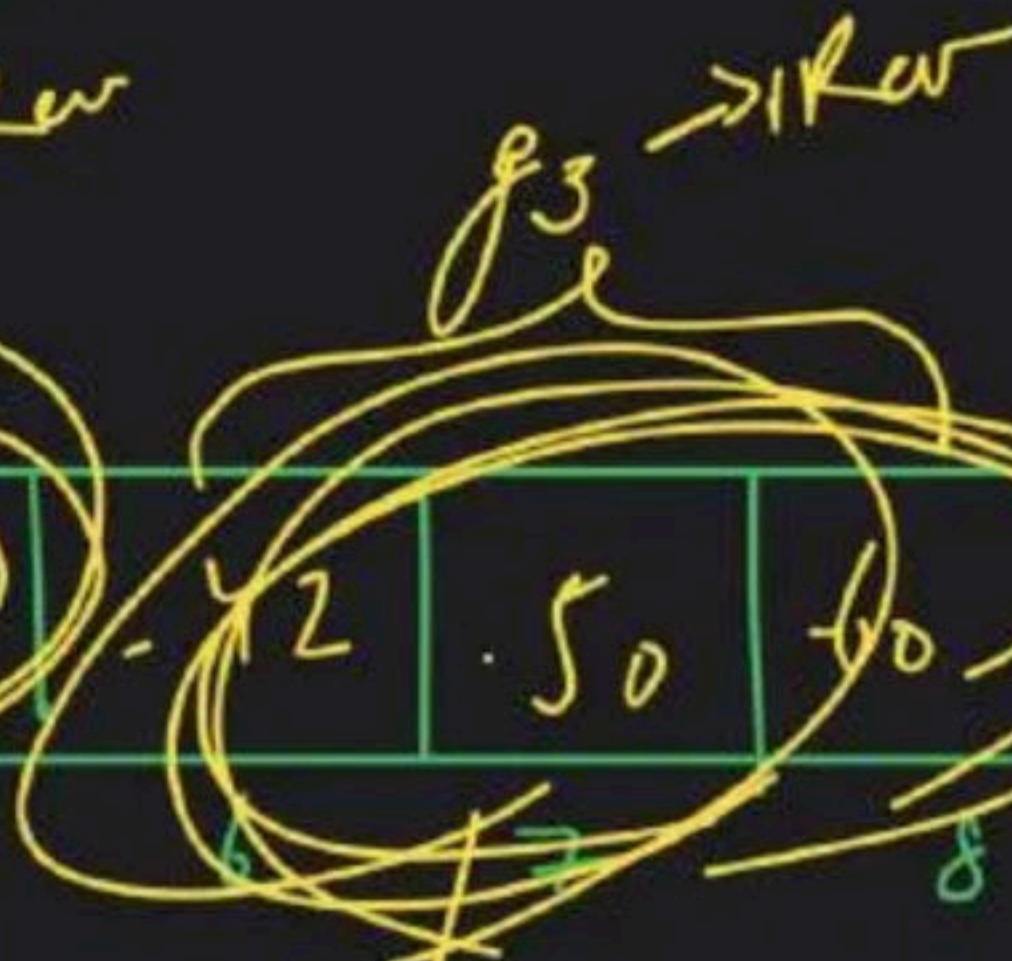
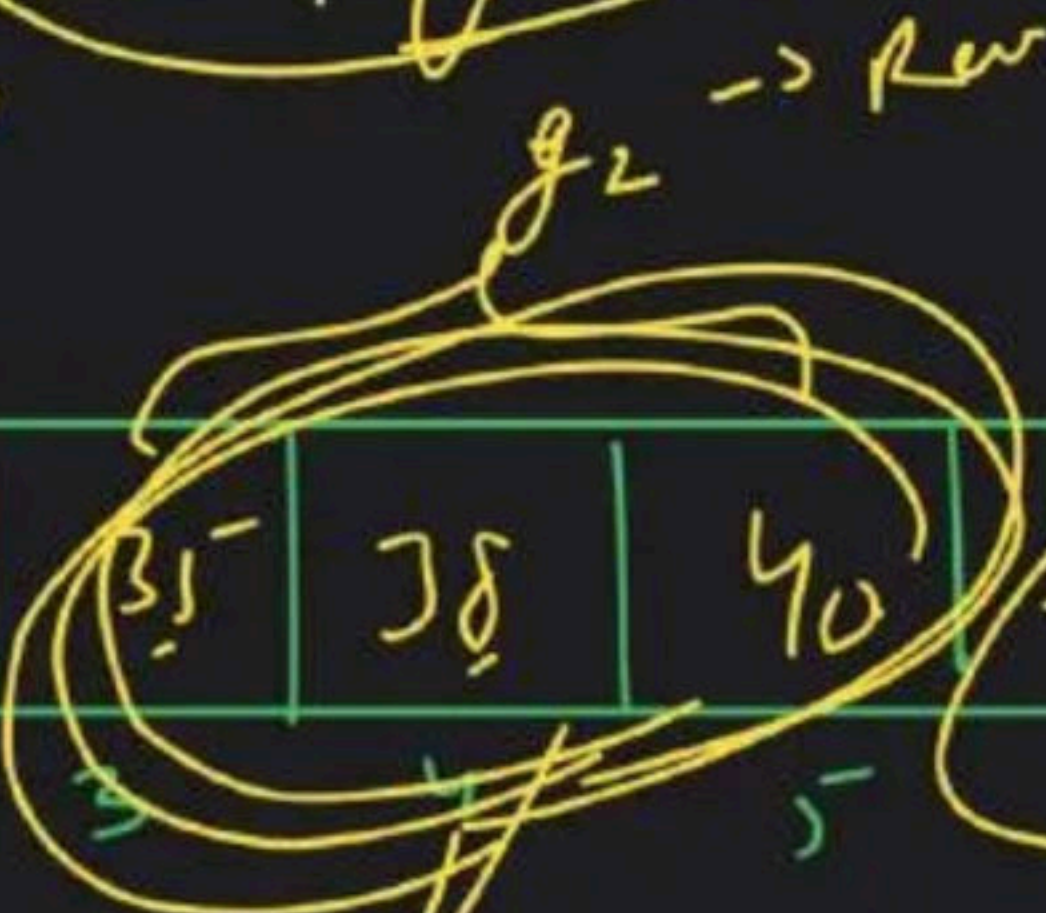
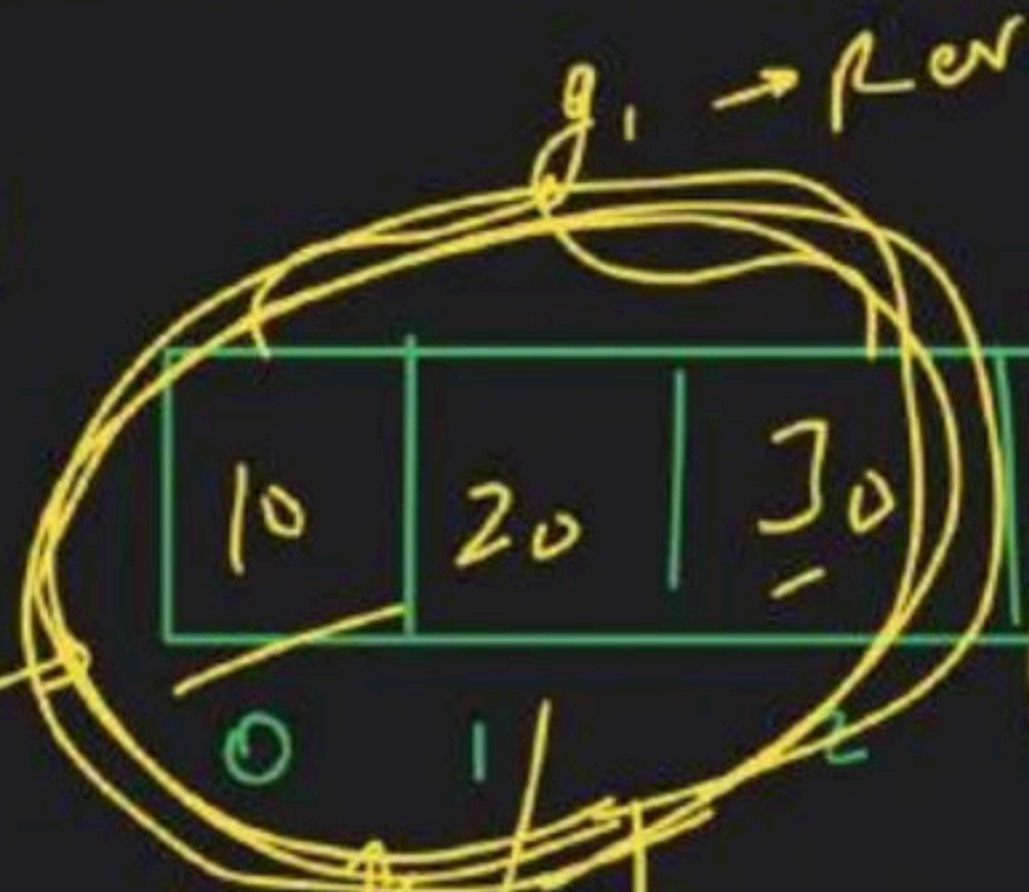




B.L

→ Reverse in K group

$K=3$



o/p

30	20	10	80	50	40	90	80	70	100	110
----	----	----	----	----	----	----	----	----	-----	-----

→ Queue

→ Pattern

3 step

2 minute

size = 11

10	20	30	40	50	60	70	80	90	100	110	30	20	10
---------------	---------------	---------------	----	----	----	----	----	----	-----	-----	----	----	----

2

(K)

11 - 3 → 8

40	50	60	70	80	90	100	110	30	20	10	60	50	40
---------------	---------------	---------------	----	----	----	-----	-----	----	----	----	----	----	----

8 - 3 → 5

70	80	90	100	110				90	80	70
----	----	----	-----	-----	--	--	--	----	----	----

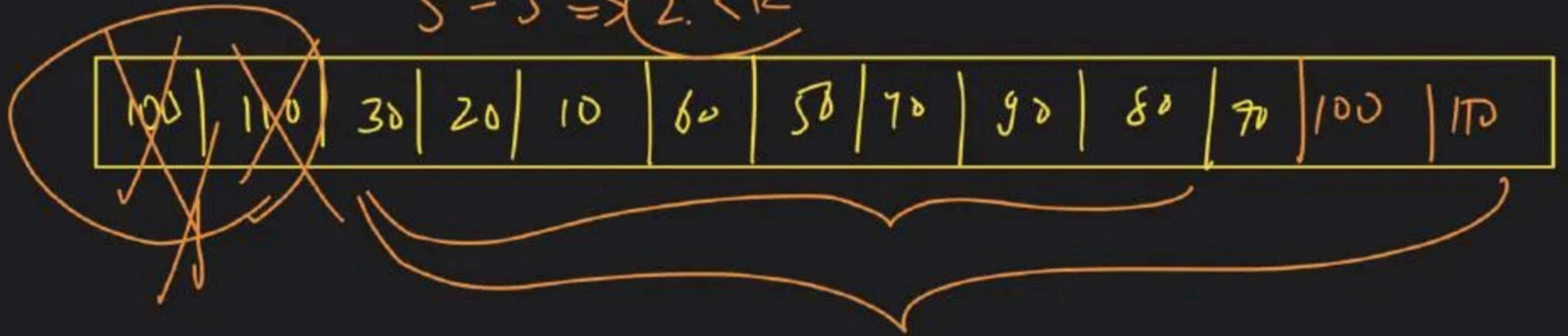
30
20
10

60
50
40

90
80
70

$$5 - 3 \Rightarrow 2 < 12$$

100	100	30	20	10	60	50	70	90	80	70	100	170
----------------	----------------	----	----	----	----	----	----	----	----	----	-----	-----



$$k=3 \quad \text{size} \Rightarrow 11$$

10
20
30
40
50
60
70
80
90
100
110

Diagram showing the first array with elements 10 through 110. The first three elements (10, 20, 30) are crossed out. A bracket labeled 'st' spans from the first element to the third element. A large bracket on the right side of the array indicates the total size of 11 elements.

$$\text{size} = 11 - 3 = 8$$

40
50
60
70
80
90
100
110
30
20
10

Diagram showing the second array with elements 40 through 110, followed by 30, 20, and 10. The first three elements (40, 50, 60) are crossed out. A bracket labeled 'st' spans from the first element to the third element. A large bracket on the right side of the array indicates the total size of 8 elements.

$$\text{size} = 8 - 3 = 5$$

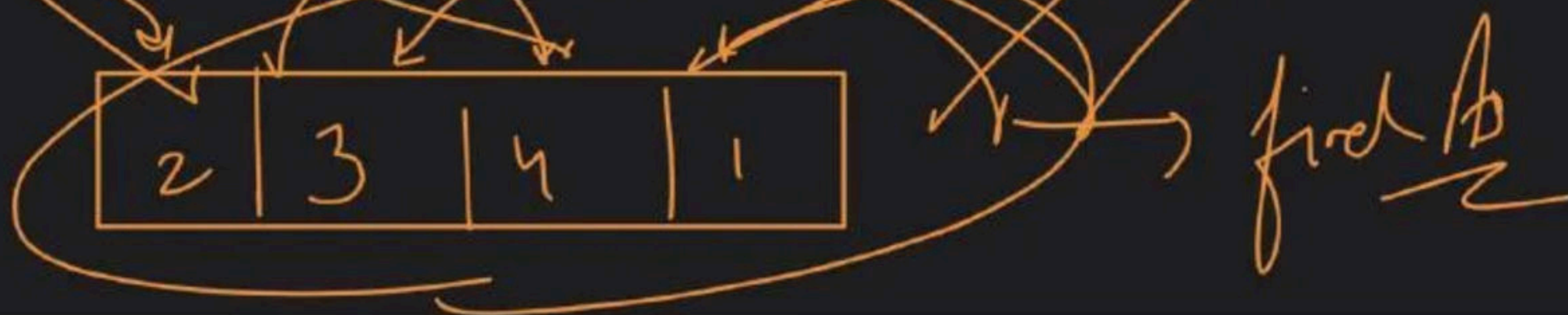
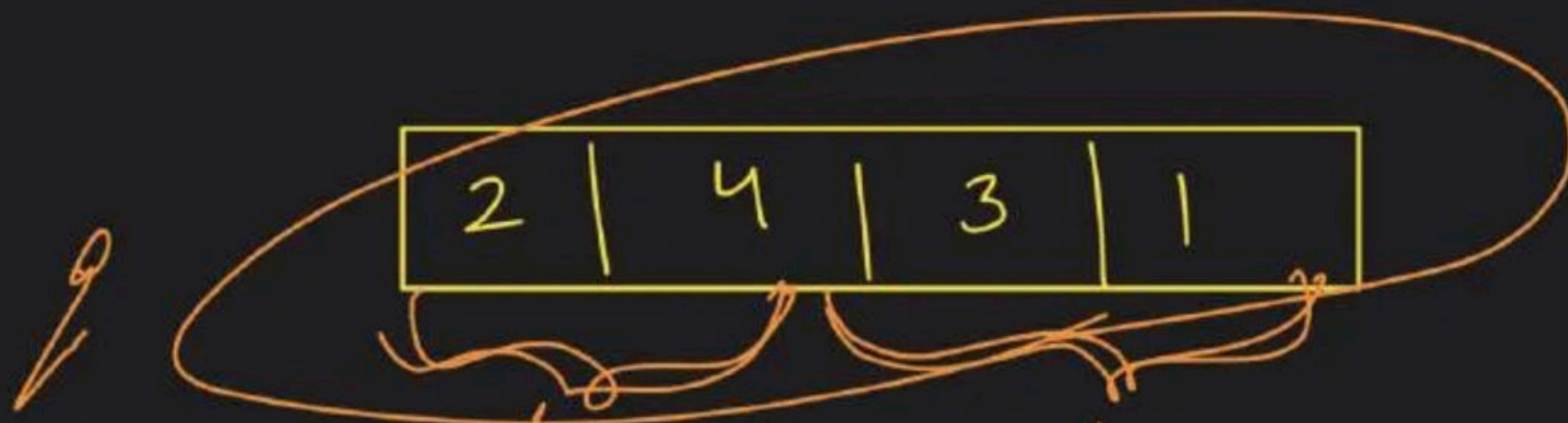
70
80
90
100
110
30
20
10
60
50
40

Diagram showing the third array with elements 70 through 110, followed by 30, 20, 10, 60, 50, and 40. The first three elements (70, 80, 90) are crossed out. A bracket labeled 'st' spans from the first element to the third element. A large bracket on the right side of the array indicates the total size of 5 elements.

$$\text{size} = 5 - 3 = 2 < k$$

100
110
30
20
10
60
50
40
90
80
70

Diagram showing the fourth array with elements 100 through 110, followed by 30, 20, 10, 60, 50, 40, 90, 80, and 70. The first two elements (100, 110) are crossed out and circled. A bracket labeled 'st' spans from the first element to the second element. A large bracket on the right side of the array indicates the total size of 2 elements. An arrow labeled 'rm' points to the circled elements.

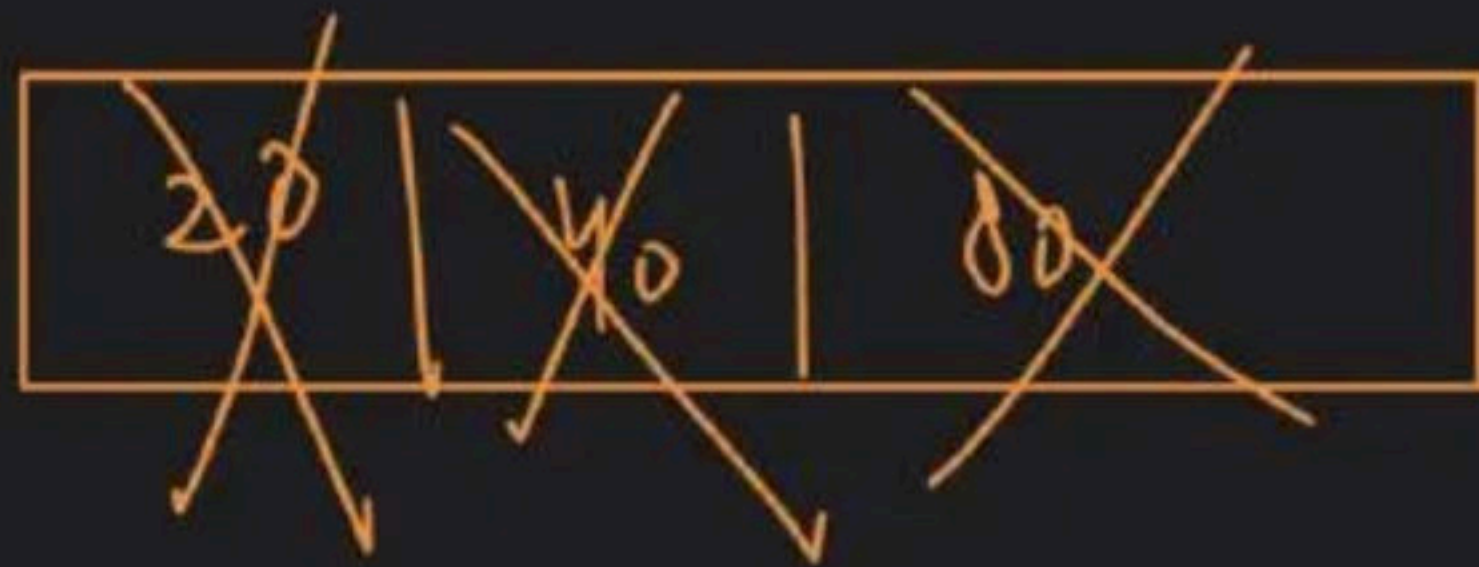


second

g₁



g₂

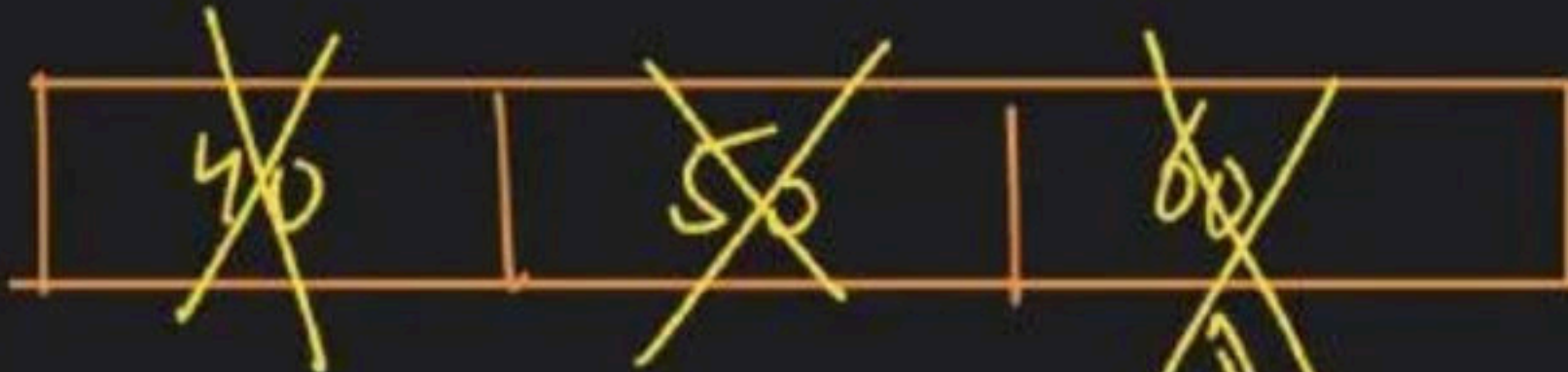


Final

Q1

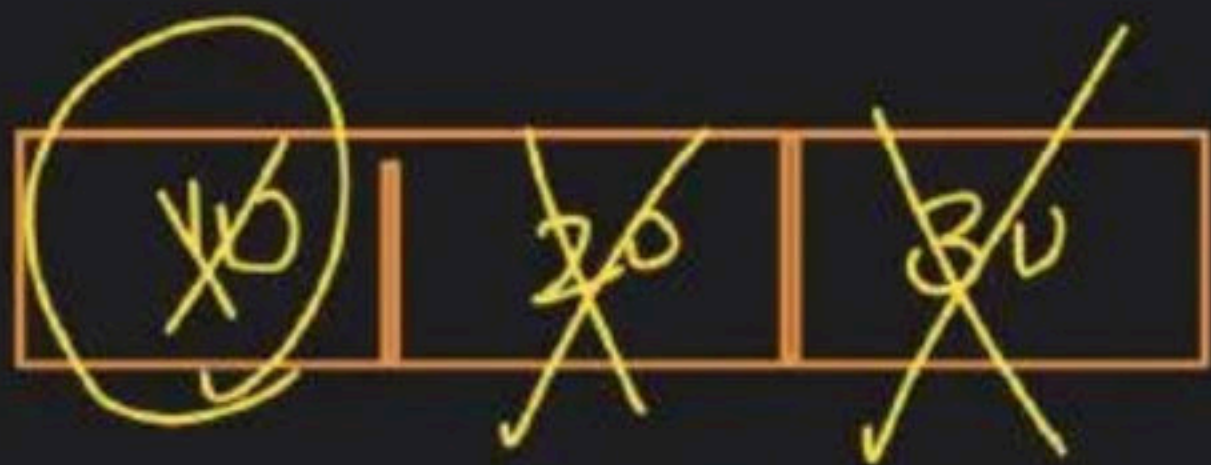
↓ step 1 → first half → queue

Q1



↗ Second half

Q2



← First Half

Ans



Range

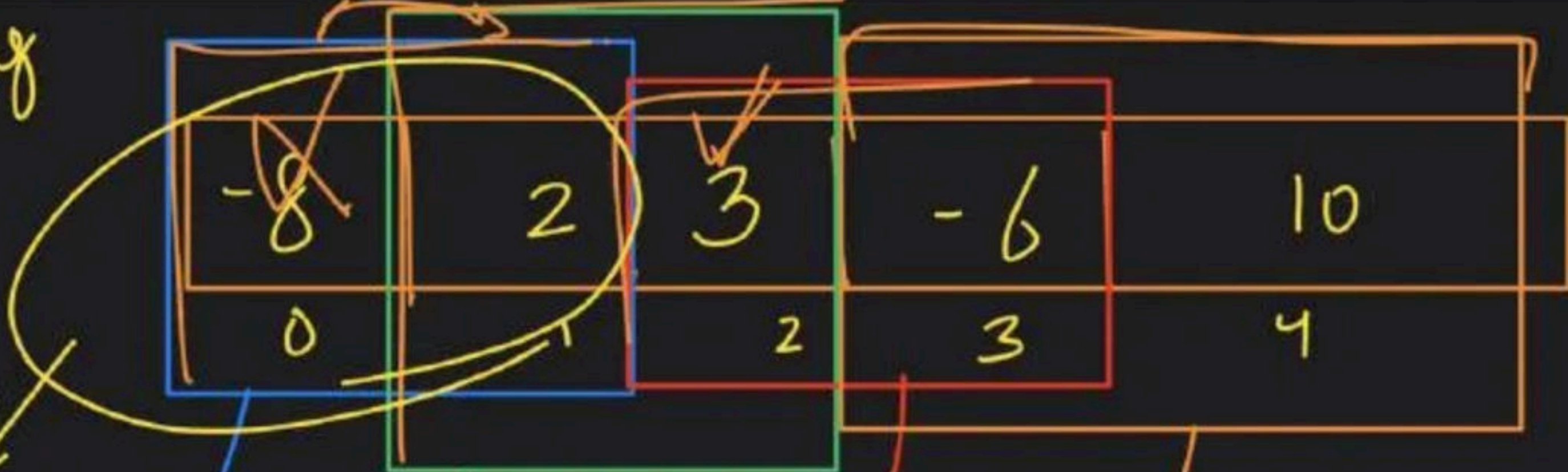
→ Sliding Window Pattern



bandwidth
size

First window

$K=2$

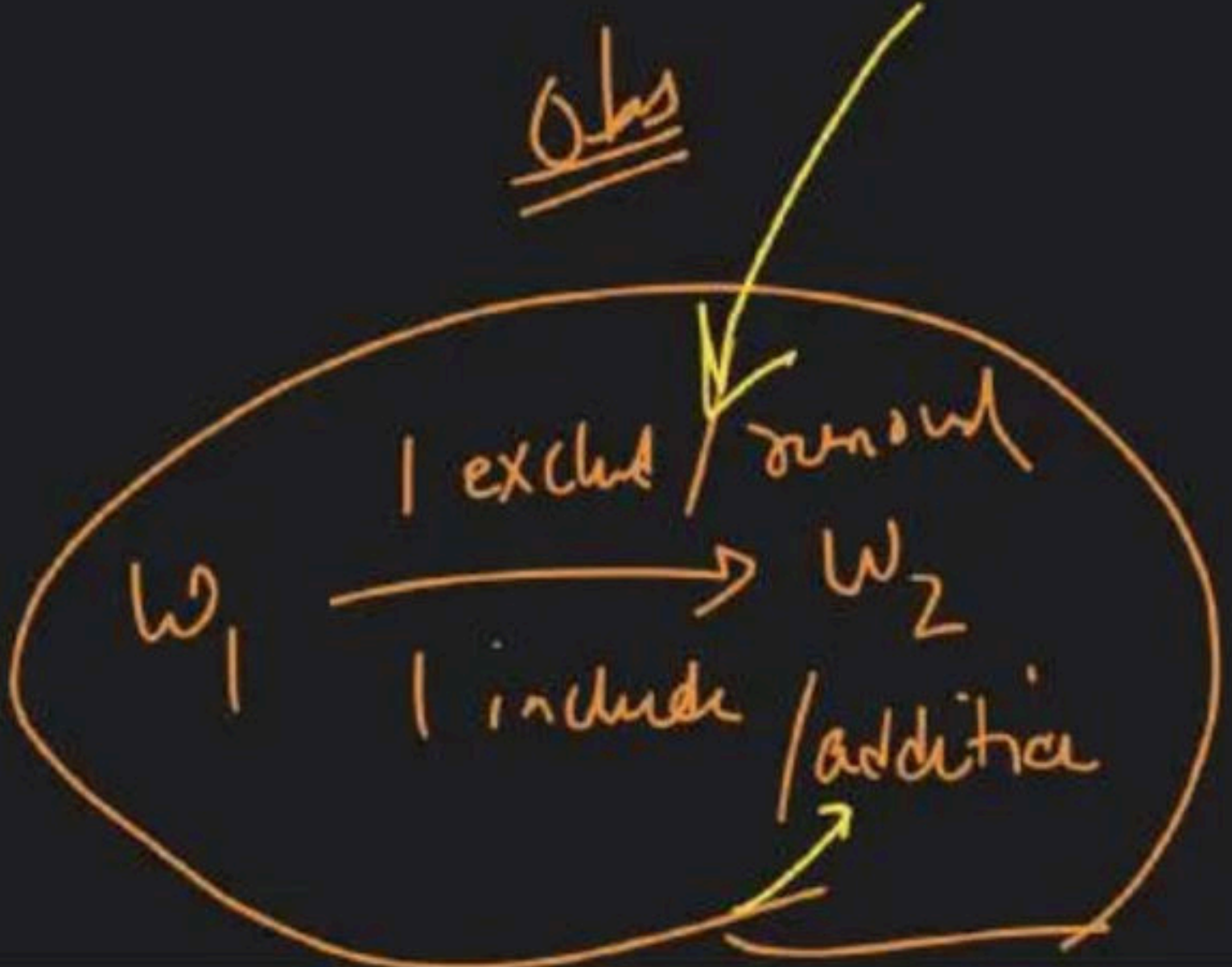


-8

0

-6

-6



3 step $\rightarrow$ SW

$\rightarrow$ ① Process First Window

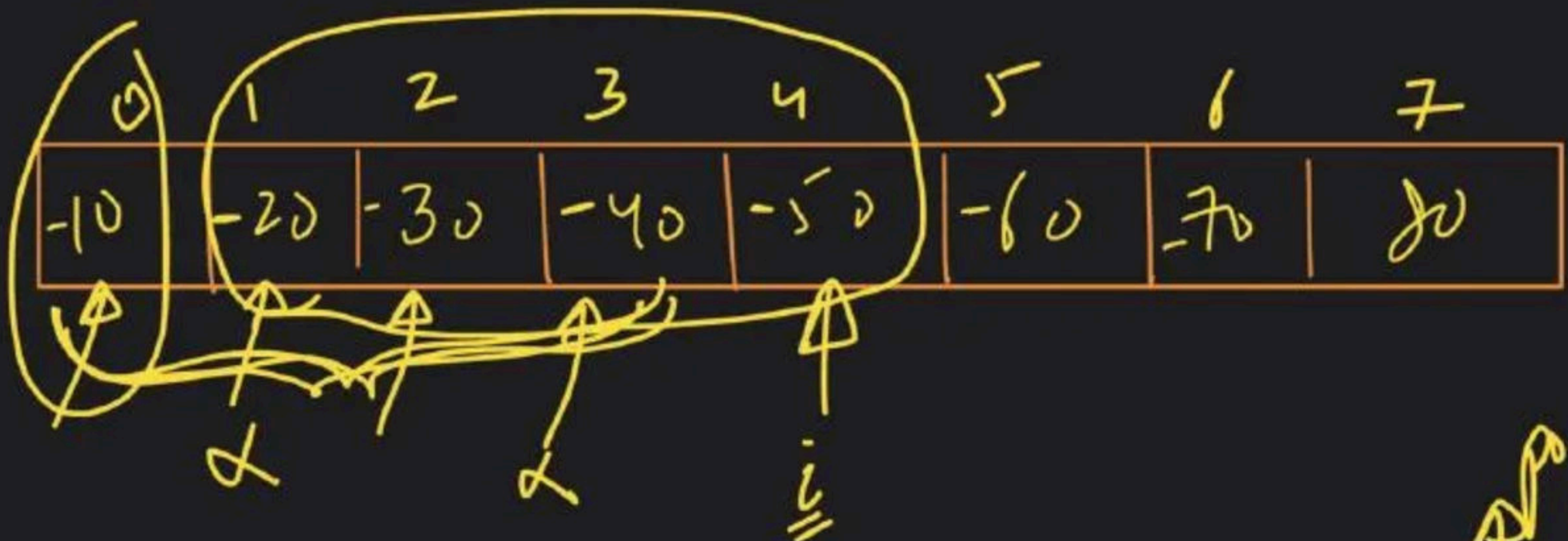
$\rightarrow$ ② Process remaining window

$\rightarrow$ ans

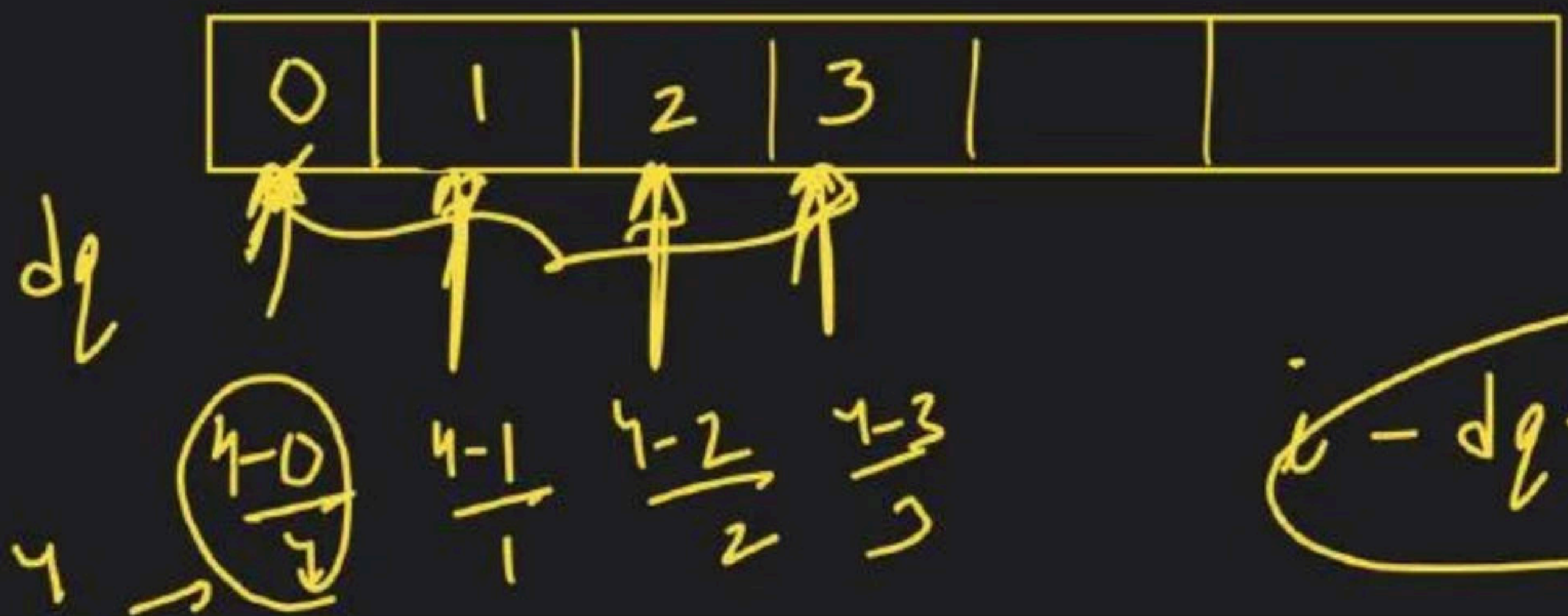
$\rightarrow$ removal

$\rightarrow$ addition

K=4

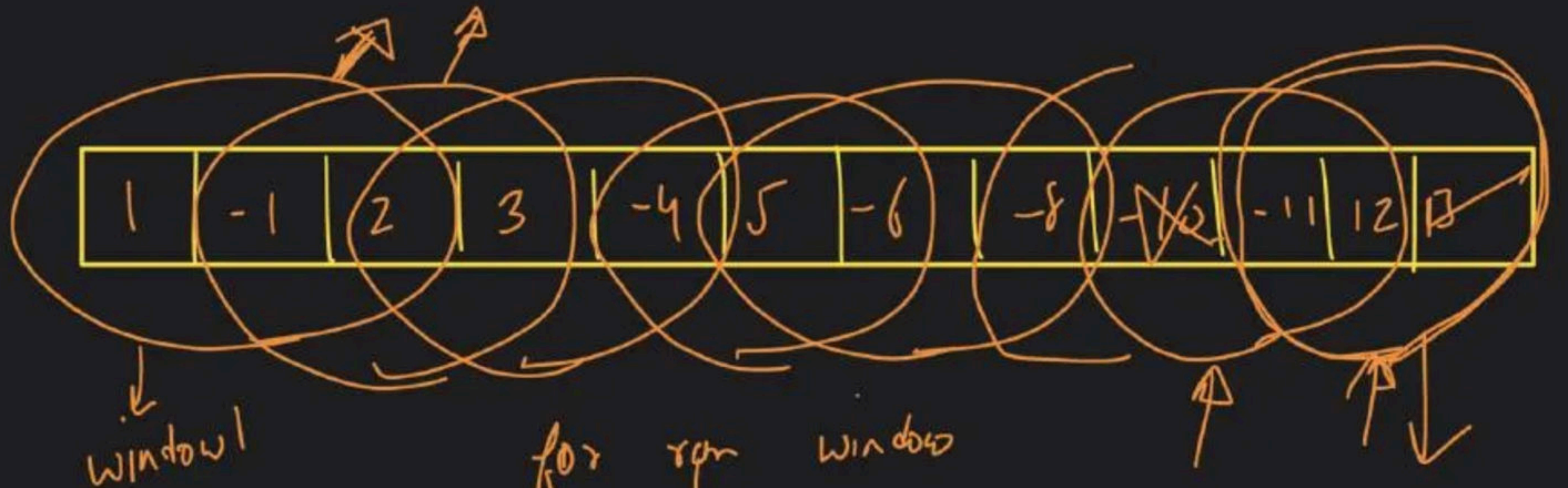


①

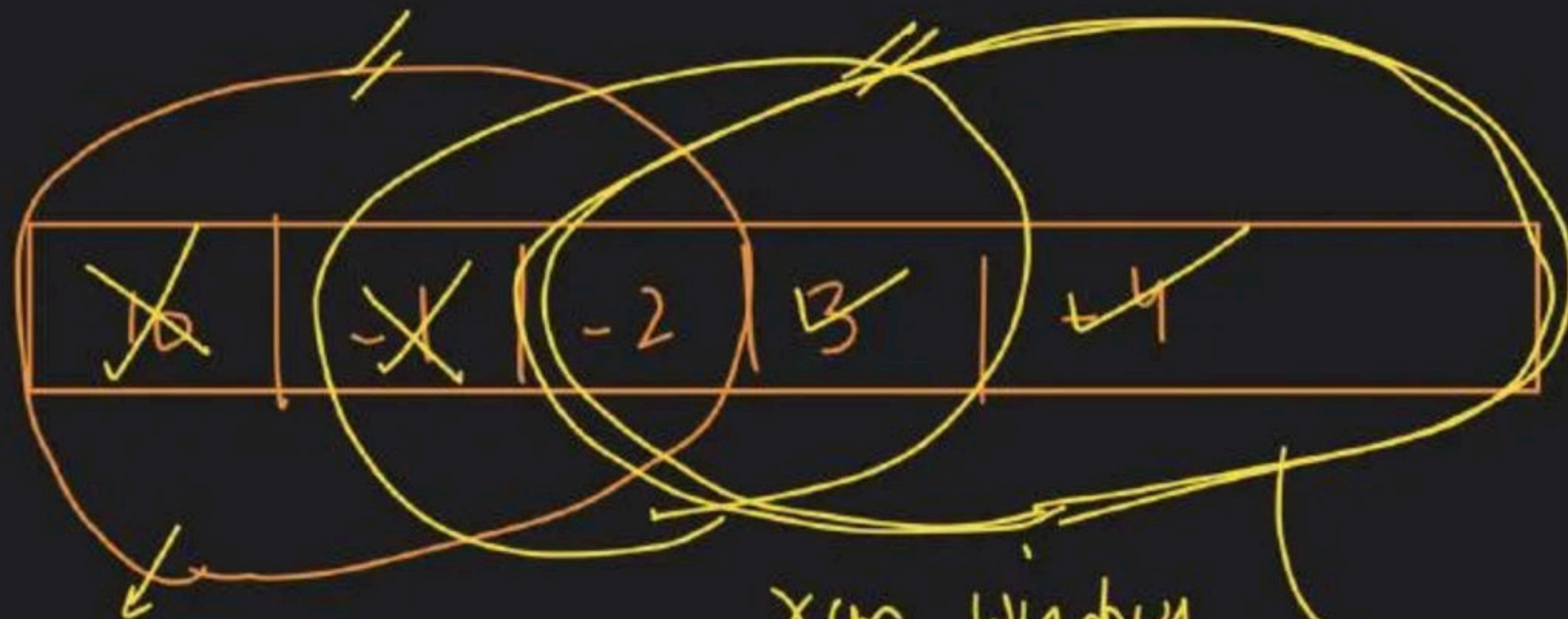


Out of range
P

$$i - dq.front() \geq K$$

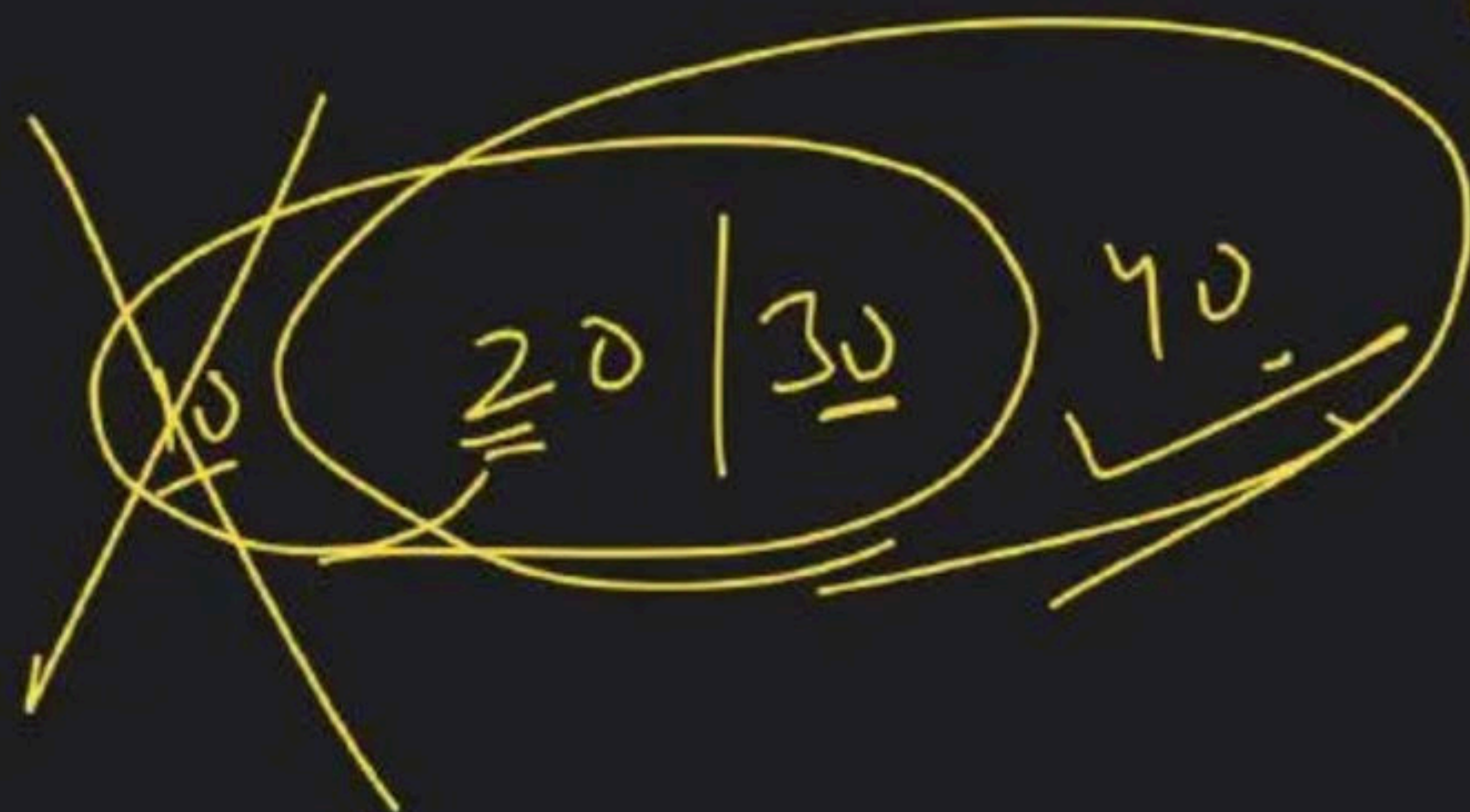


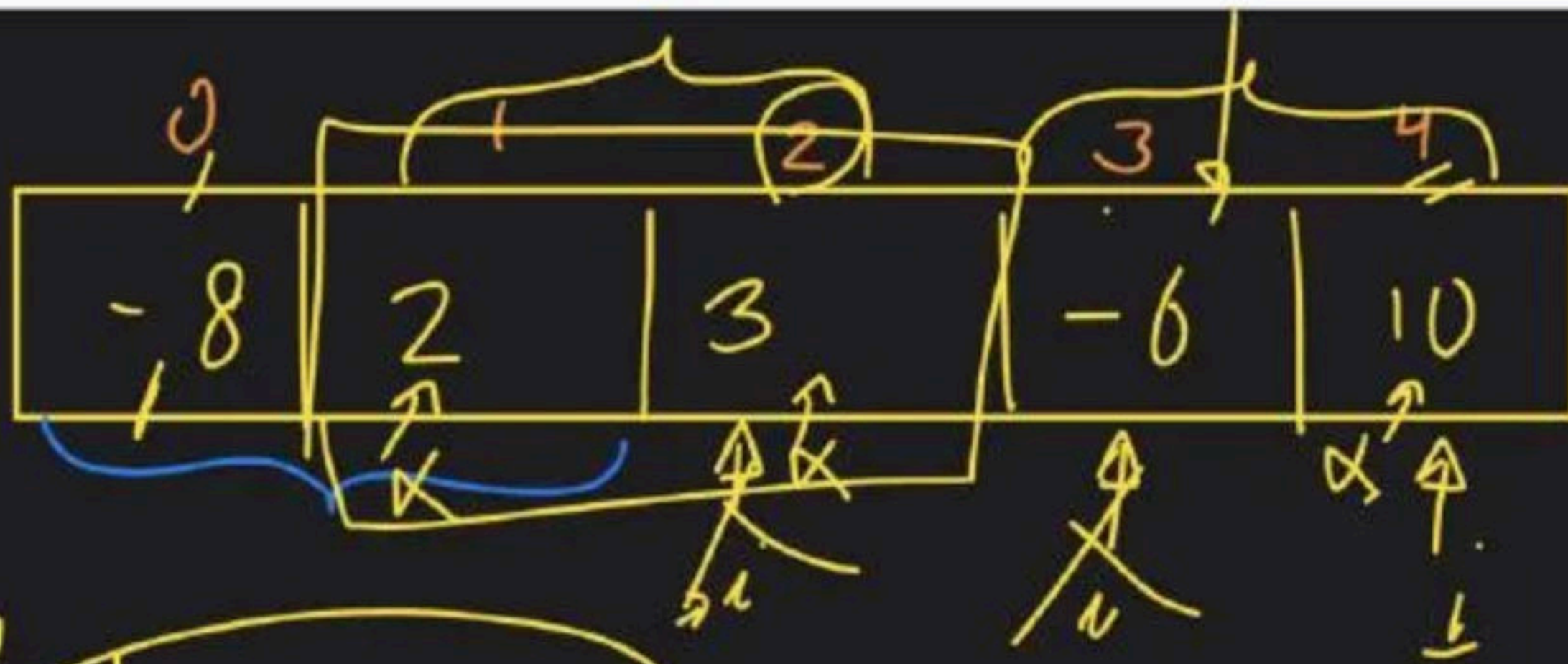
$k=3$



W_1

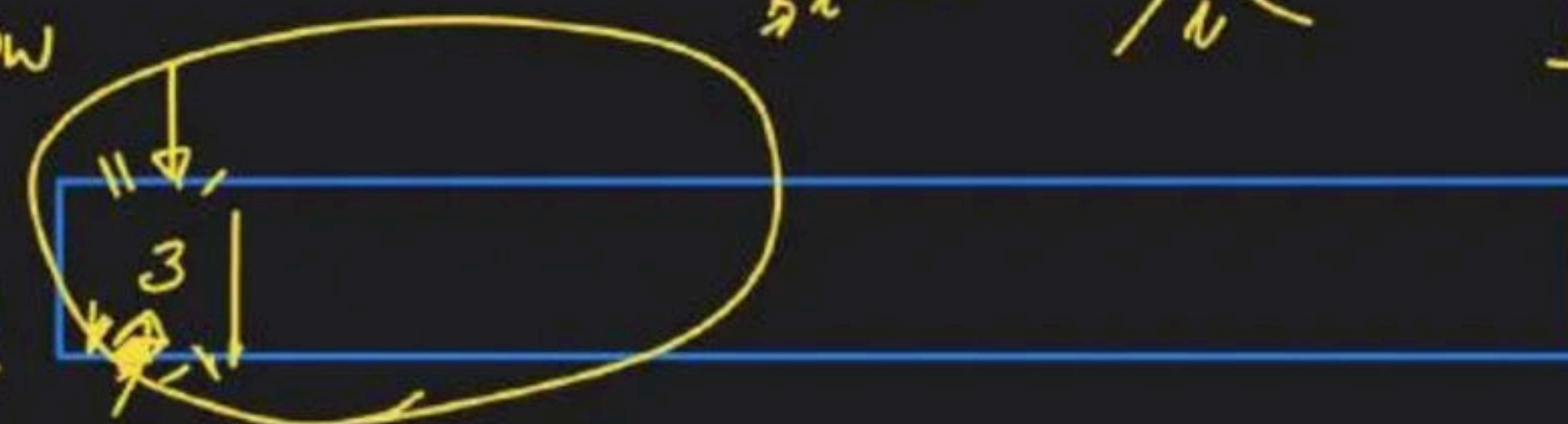
sliding window





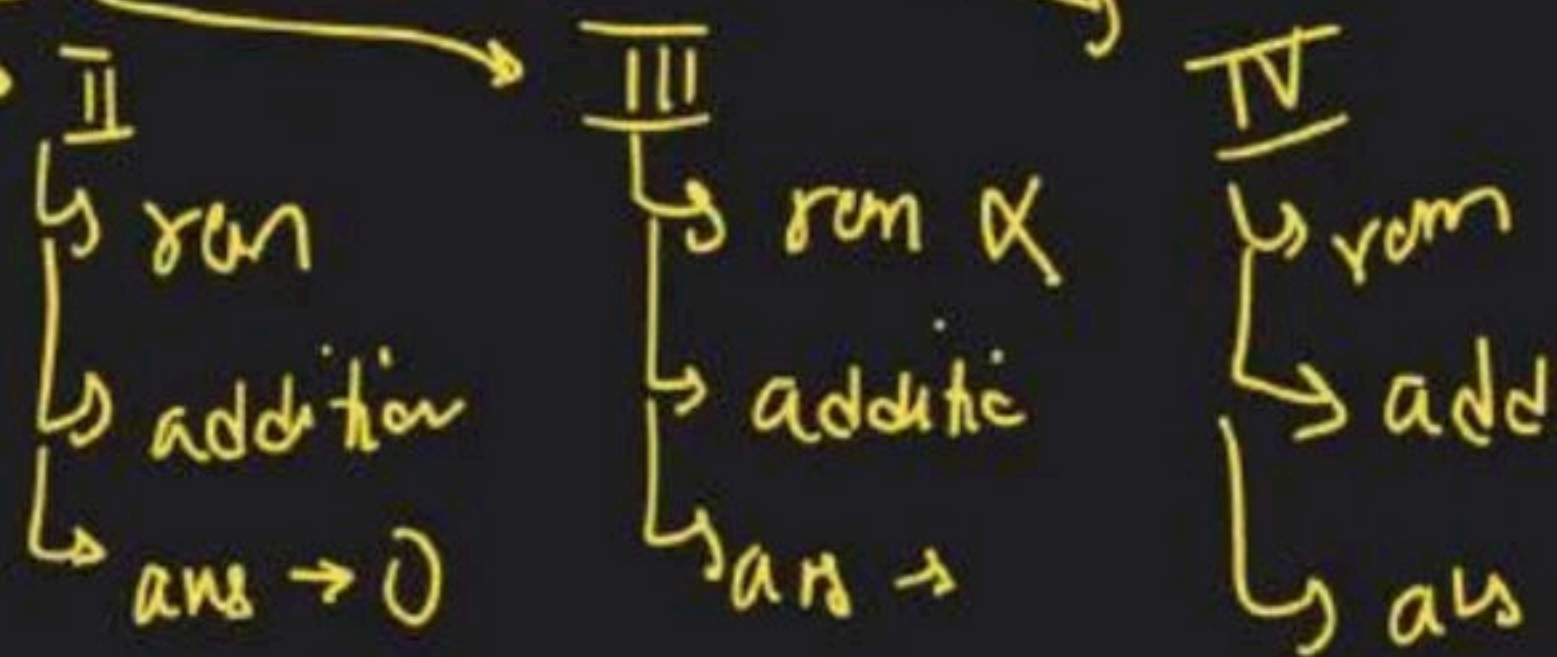
$K=2$

① Ist window



Ist window $\rightarrow$ ans

② Rem Window



81

ans

-8
0
-6
-6

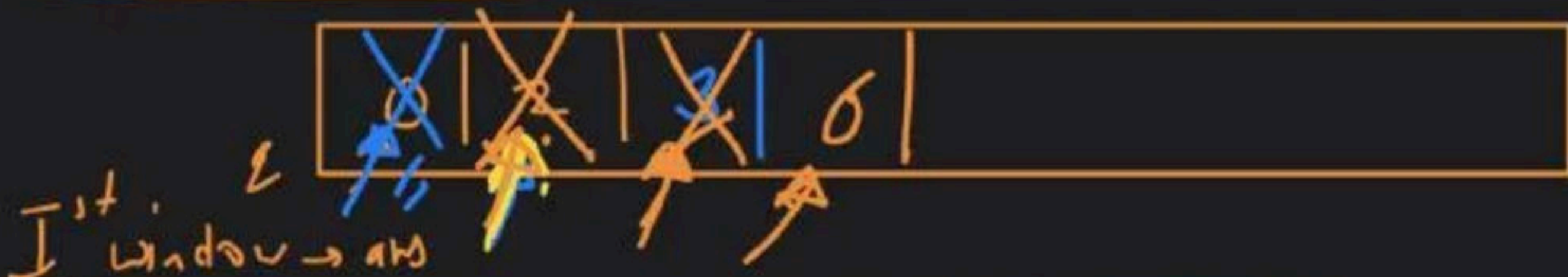
K=3



$$i - \text{left} \geq K$$

↓
out of window

1st window passes



sum window

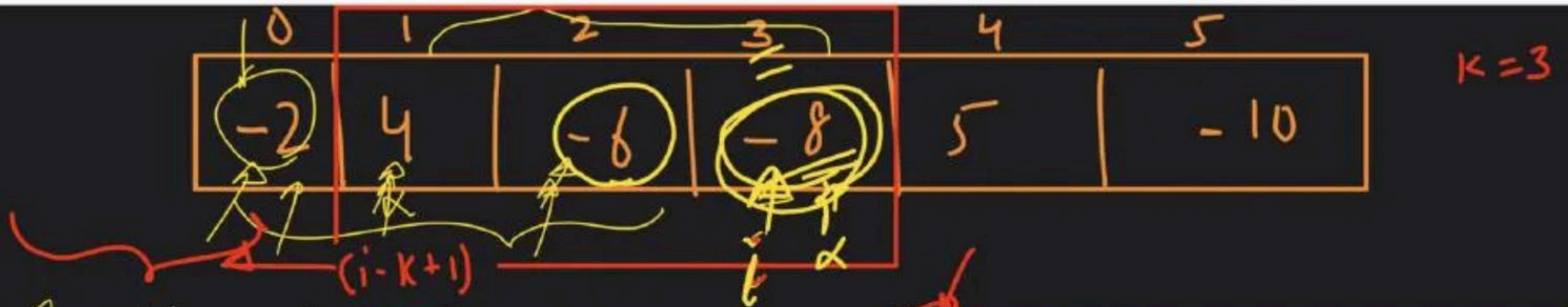
II
→ sum
→ add
→ ans

III
→ sum
→ add
→ ans

IV
→ sum
→ add
→ ans

V
→ sum
→ add
→ ans

-1
-4
-5
-10



(2) Remaining window $\rightarrow$ process



(II) $\rightarrow$ removal $\rightarrow$ current window $\rightarrow (i-K+1) \rightarrow i$

$$dq.front() < i-K+1$$

$\rightarrow$ remove krd

$\rightarrow$ addition $\rightarrow$ -ve $\rightarrow$ index jth

$\rightarrow$ ans

Kal →

Extra Class

8:30

Quick

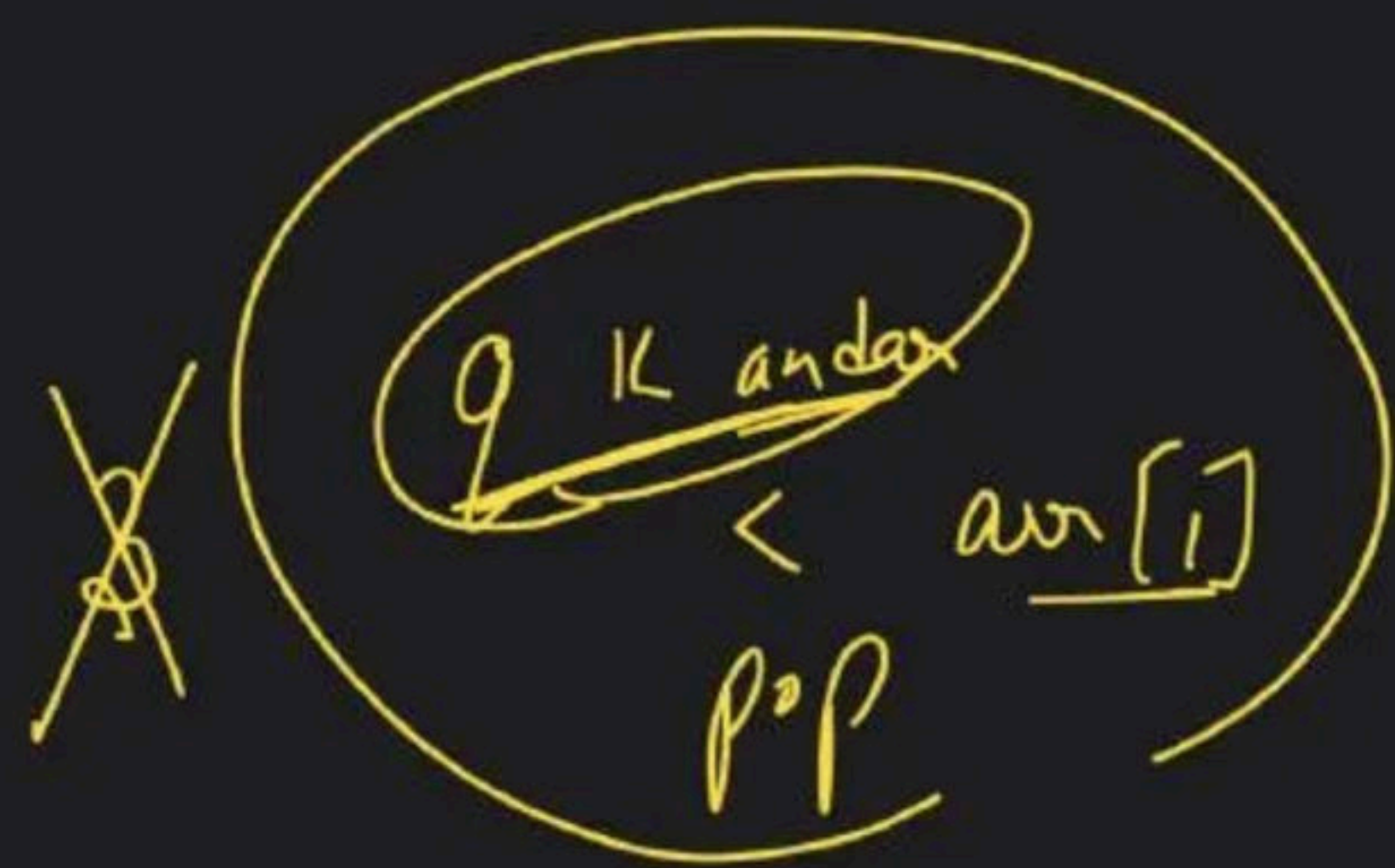
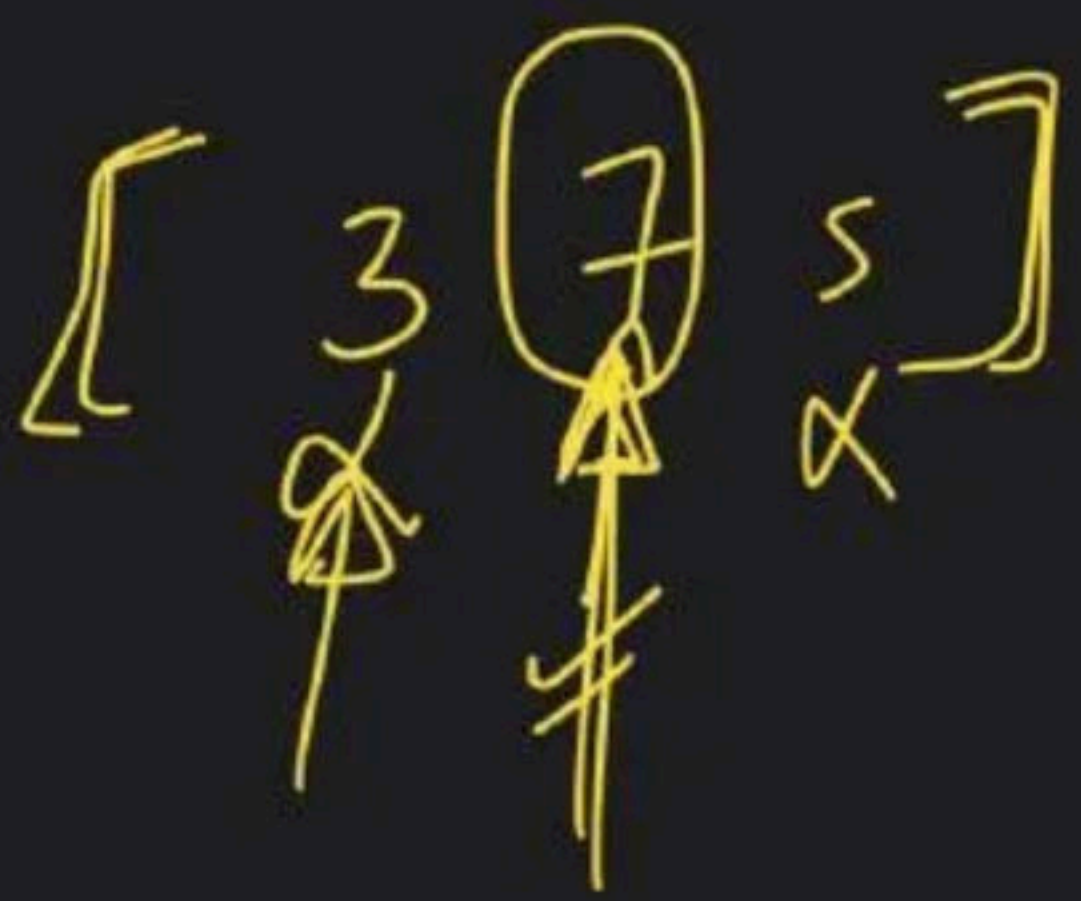
Complete

Queue Class - 3

Special class

→ Sliding Window Maximum

$\rightarrow T_s^+$
 T_s^- ay
 then
 by scan
 by add
 by ans

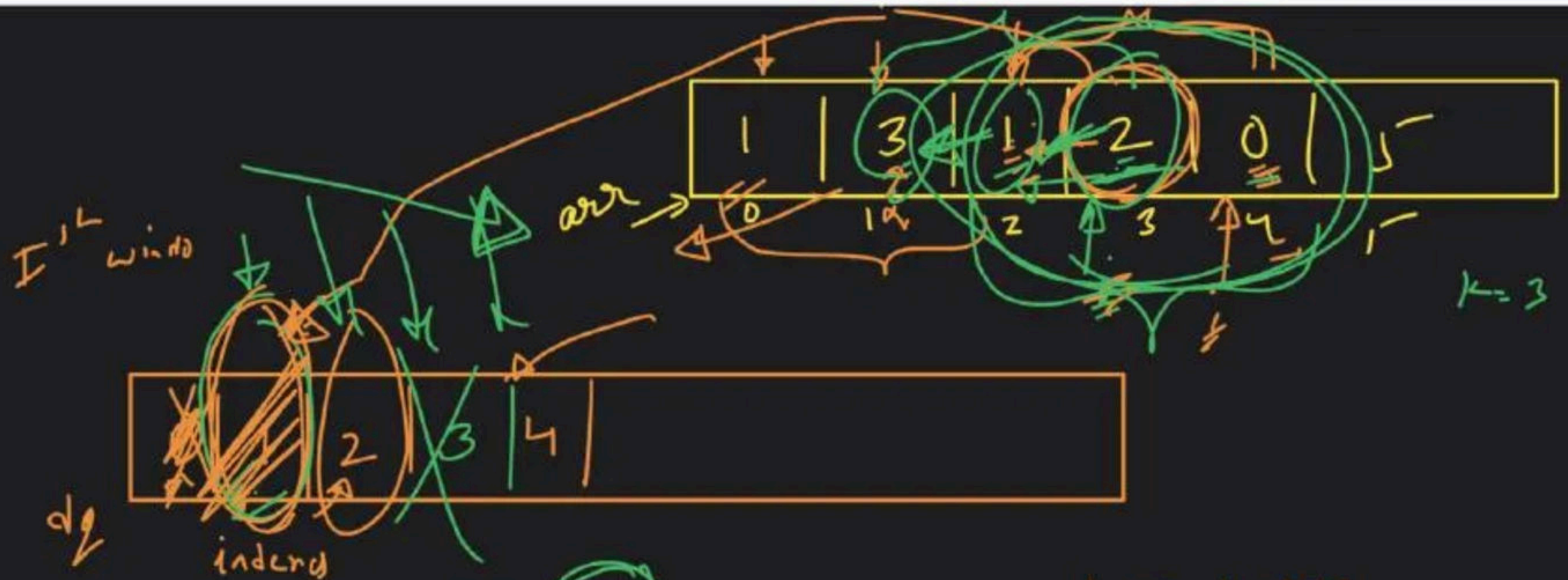


1 | 3 | -1 | -3 | -5 | 3 | 6 | 7
 0 1 2 3 4 5 6 7

$k = 3$

front index	value
7	7
Back	

3, 3, -1, 3, 6, 7



1st window → 3 3 1

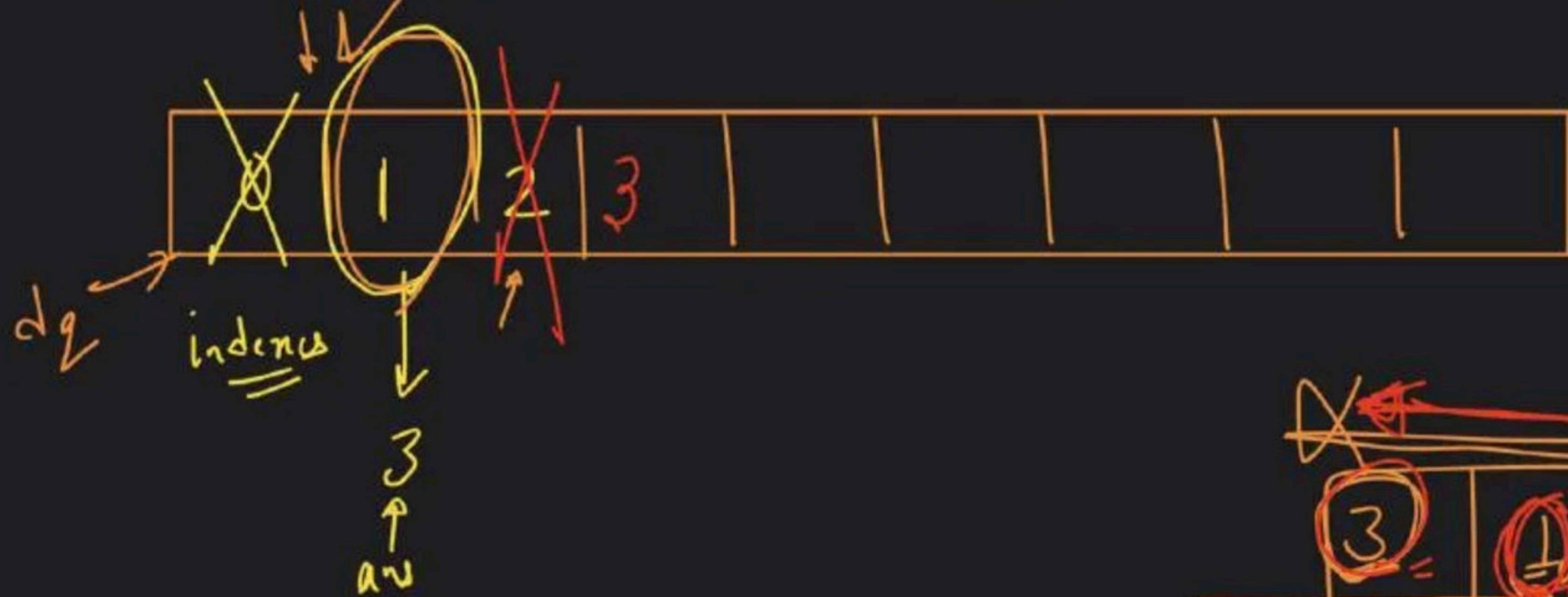
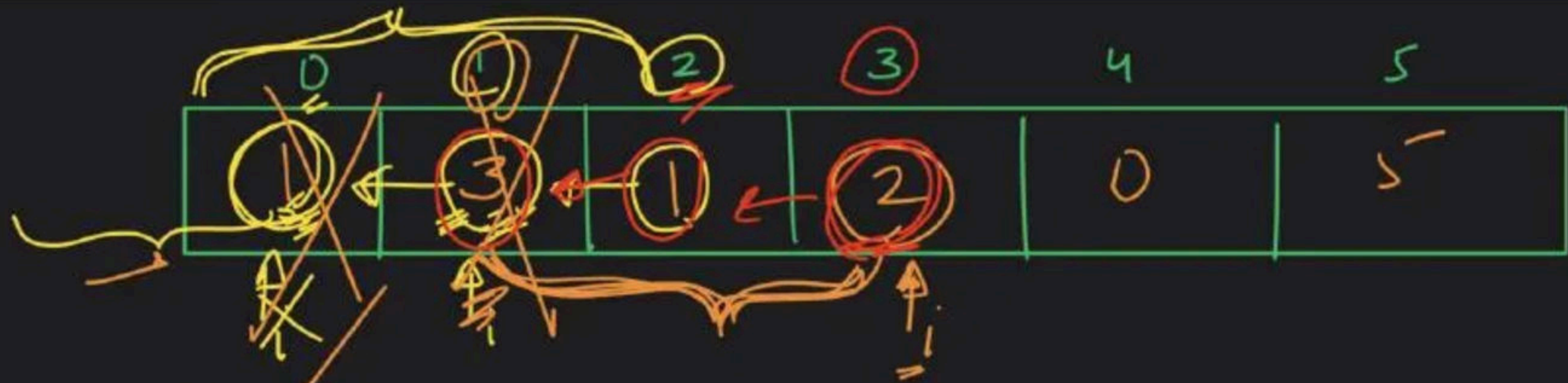
$1 < 4-3 \rightarrow T$
 $1 < 2 \rightarrow T$

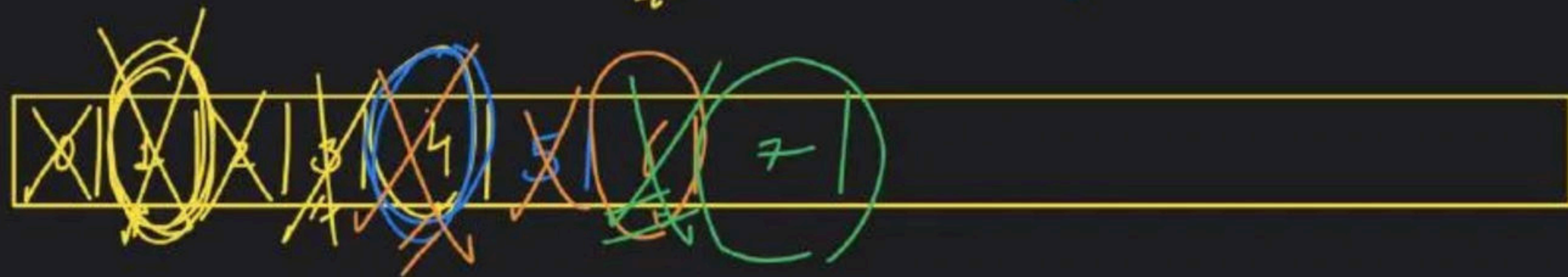
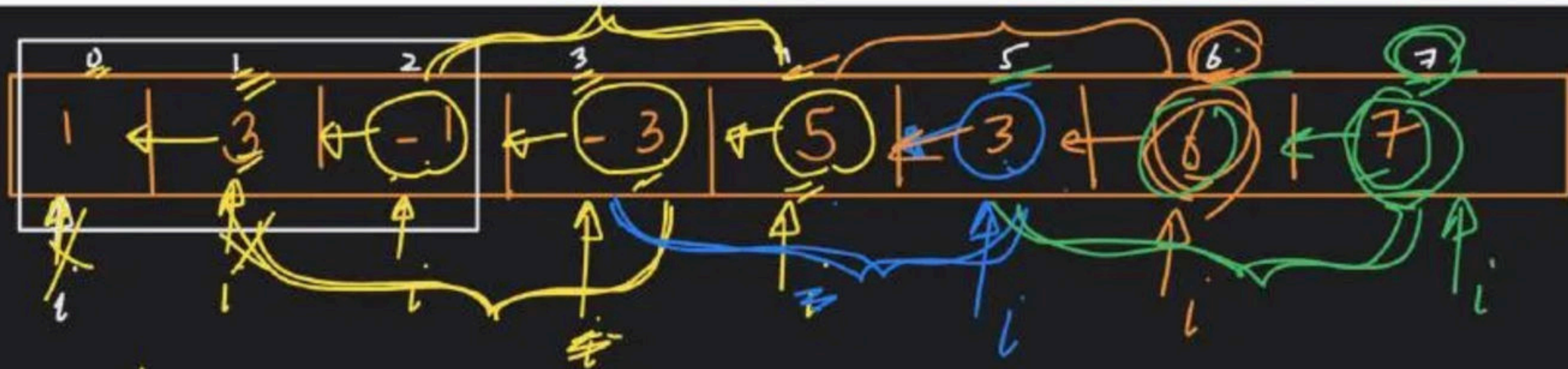
Run → II

- run
- add
- ans

III

- run
- add
- ans →





dg

$$\text{I} \rightarrow 3$$

$$\text{II} \rightarrow 3$$

$$\text{III} \rightarrow 5$$

$$\text{IV} \rightarrow 5$$

$$\text{V} \rightarrow 6$$

$$\text{VI} \rightarrow 7$$

$$\{3, 3, 5, 5, 6, 7\}$$



freq

char	int
a	2
c	1
d	2
e	1

a a c d d e

↑ ↑ ↑ ↑ ↑

i i i i i

~~a~~ | ~~a~~ | (c) | d | d | e

↑ ↑ ↑

ans → a#c#c#c

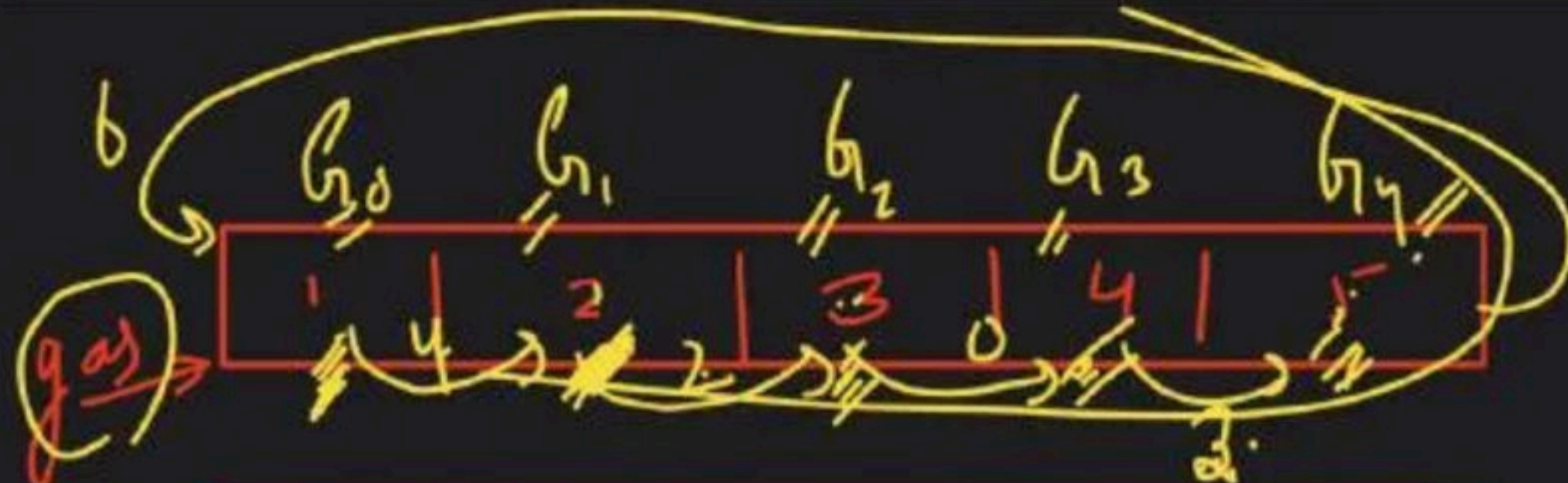
(A) $\downarrow^{\textcircled{1}} \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow$
a a b e d a b b a j a b b a

(2) a | a | b | a | d | a | b | b | a | j | a | b | b | a |

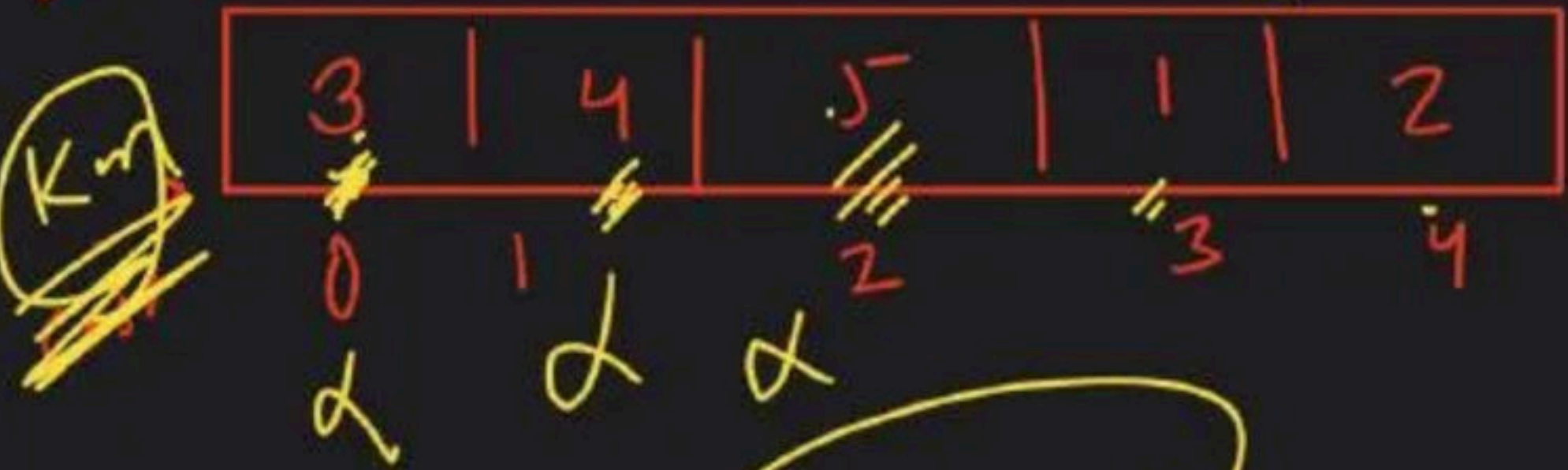
↑ (3) ↑ ↑ ↑ ↑

ans → a # b b b b d d d d d d d

char	int
(3) a	7
b	5
d	1
j	1



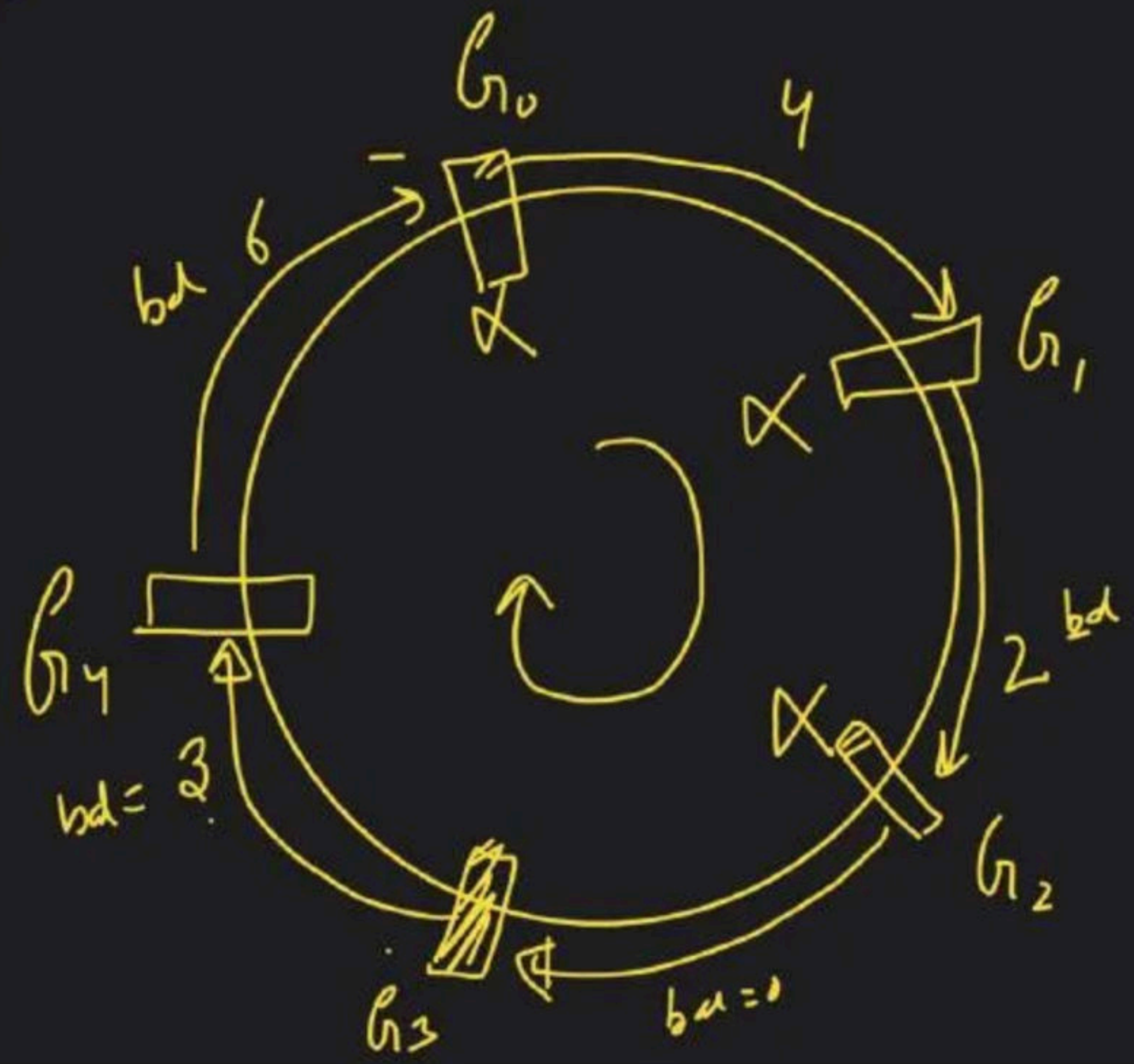
1 litre gas $\rightarrow$ 1 km



ans = 3
index

Route
force

$O(n^2)$



Car =

1 | 2 | 3 | 4 | 5

(M1) $\rightarrow$ Brute

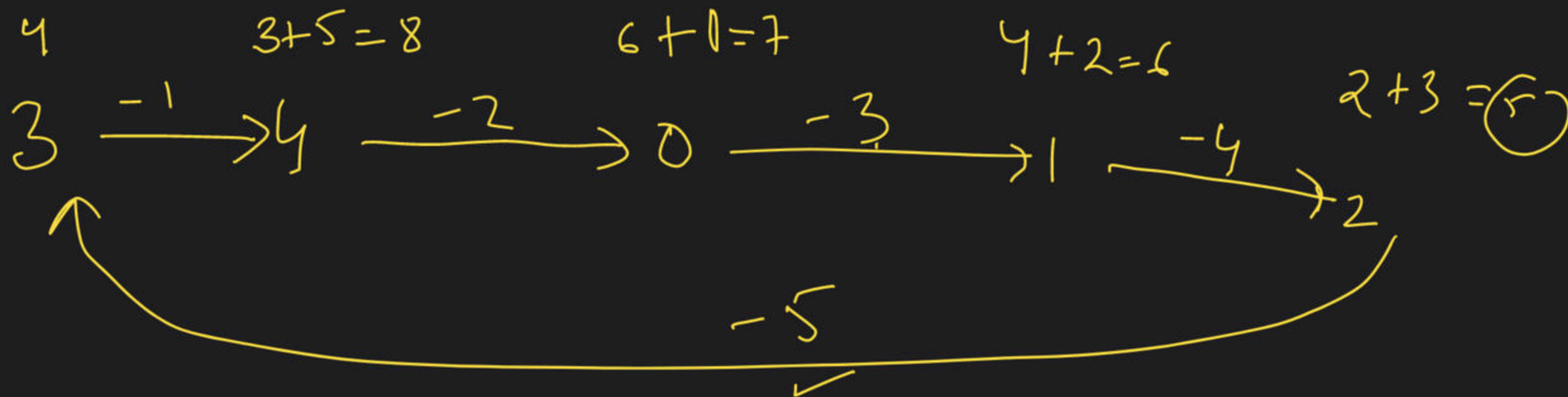
Cost =

3 [4 | 5 | 1 | 2]



Simulation $\xrightarrow{\text{even index}} \mathcal{O}(n^2)$

Reach

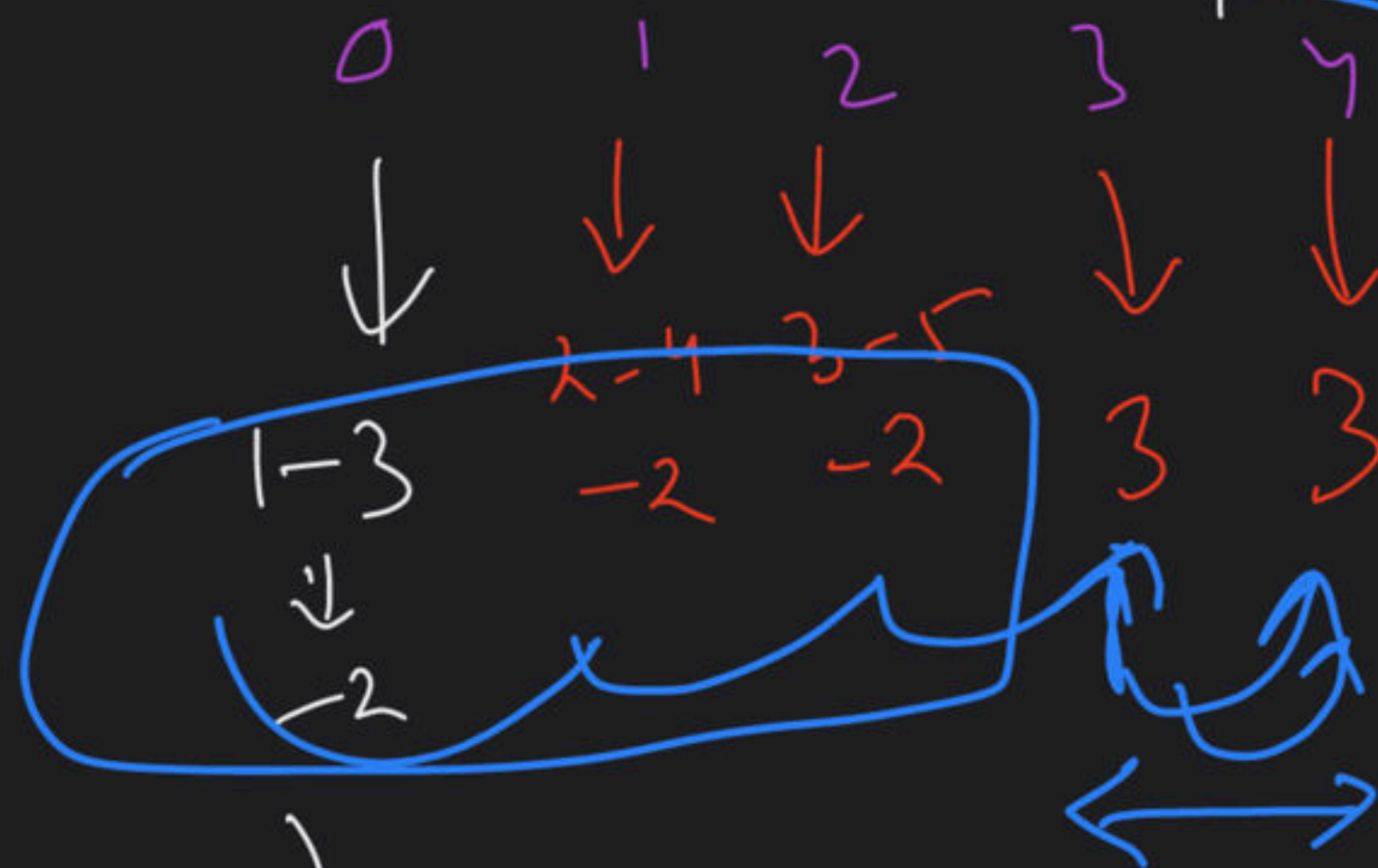


Car 2

1	2	3	4	5
3	4	5	1	2

$$4 - 1 = 3 + 5$$

$$8 - 2 \Rightarrow 6$$



$$\text{Surplus} = 3 + 3 = 6$$

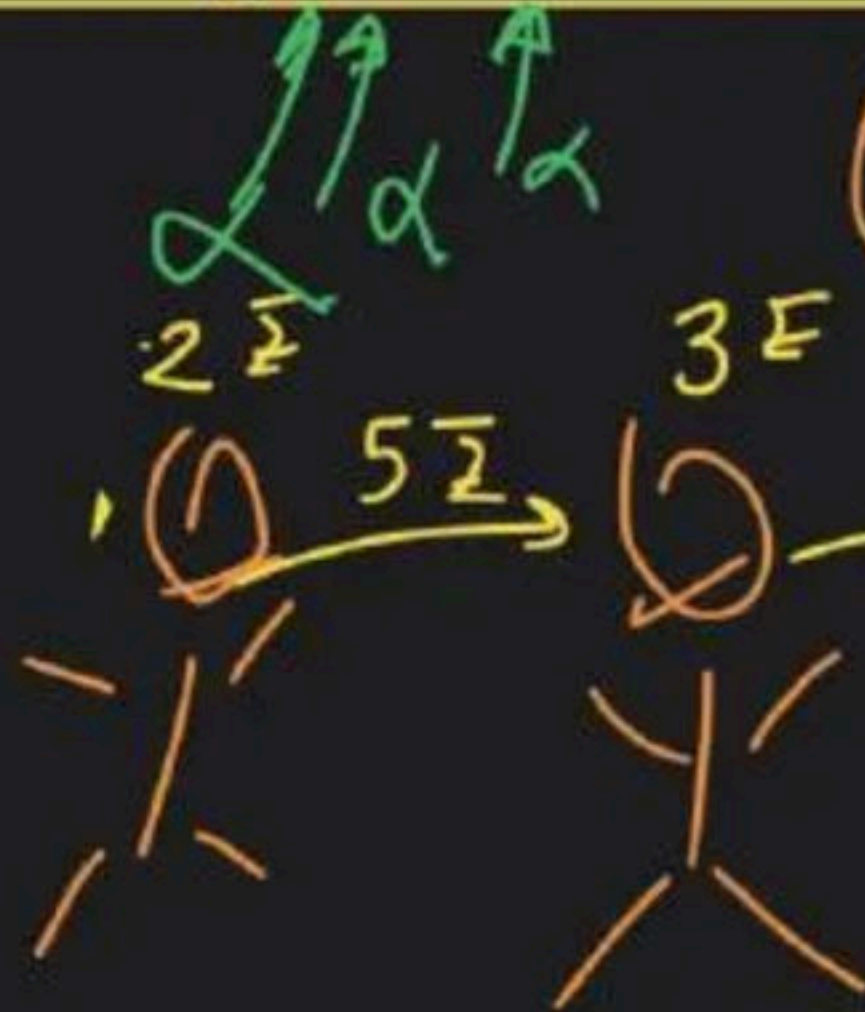
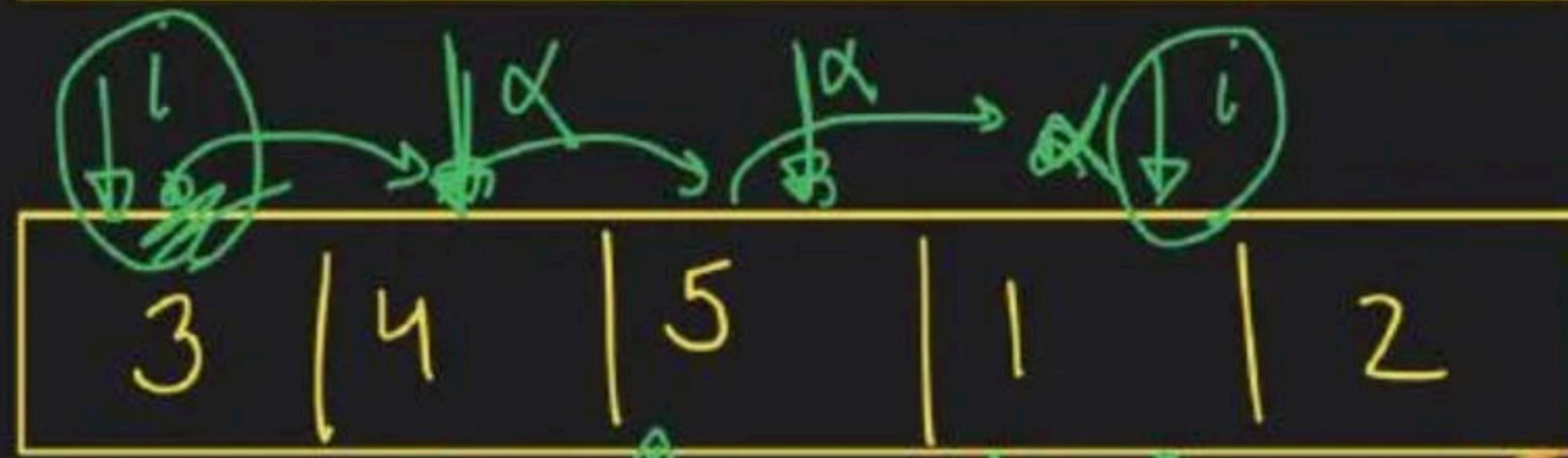
$$\text{deficit} = 6$$

$$\text{start} = 3$$

if (surplus < 0)

- deficit += (-surplus)
- start = i + 1
- surplus = 0

$$\text{Surplus} += \text{gas}[i] - \text{cost}[i]$$



$8 - i$

$8 \pm$



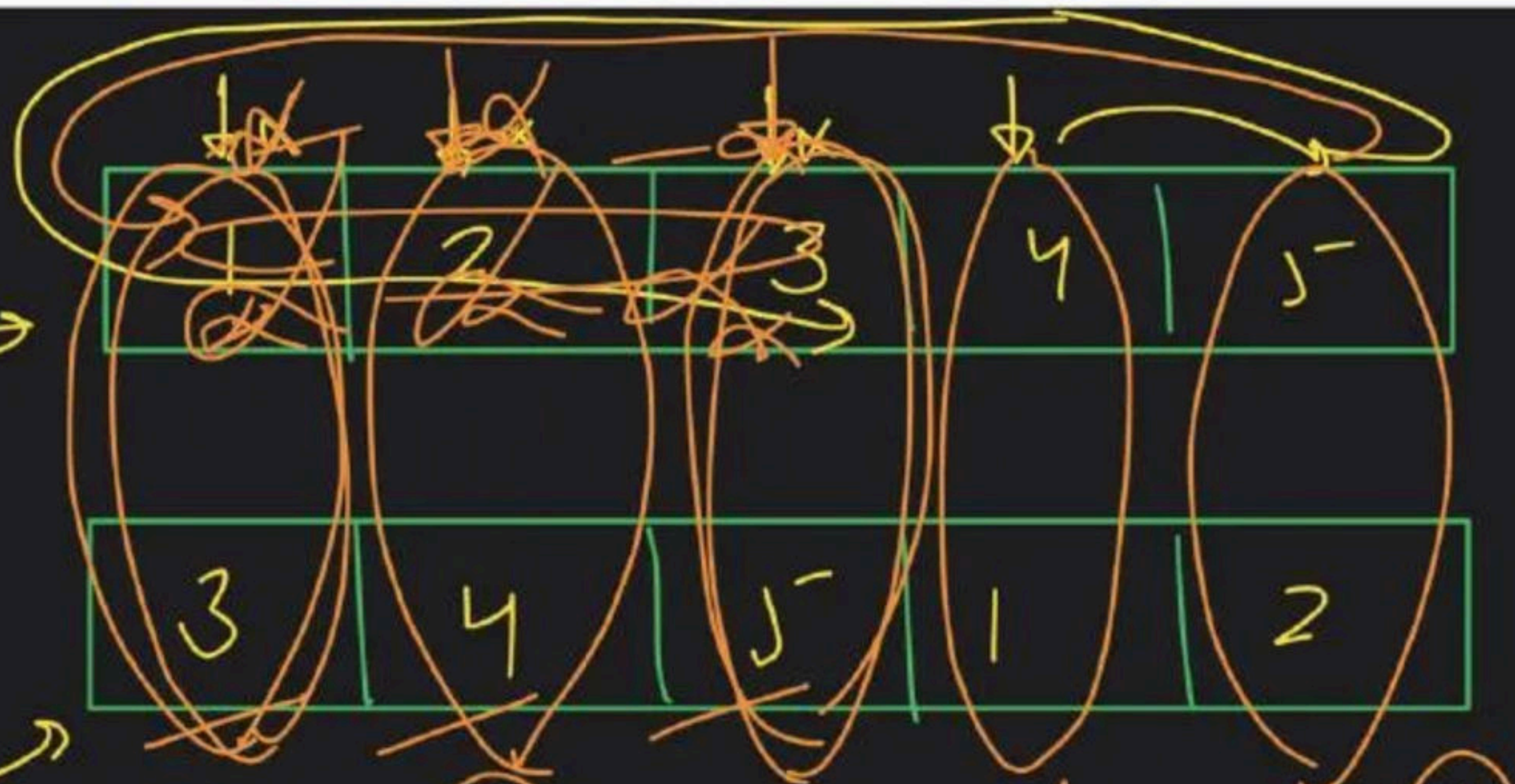
$10 \pm$ need

Why?

i

gas →

km →



2
deficit

2
deficit

2
deficit

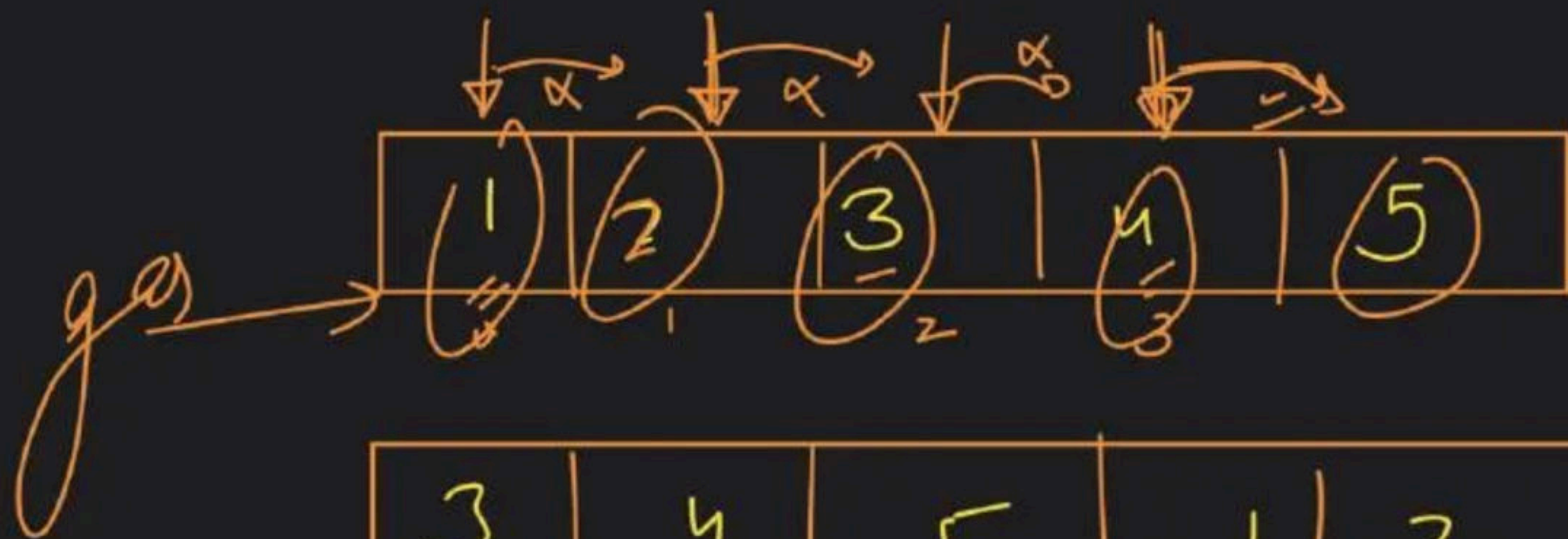
bal = 3

bal = 1

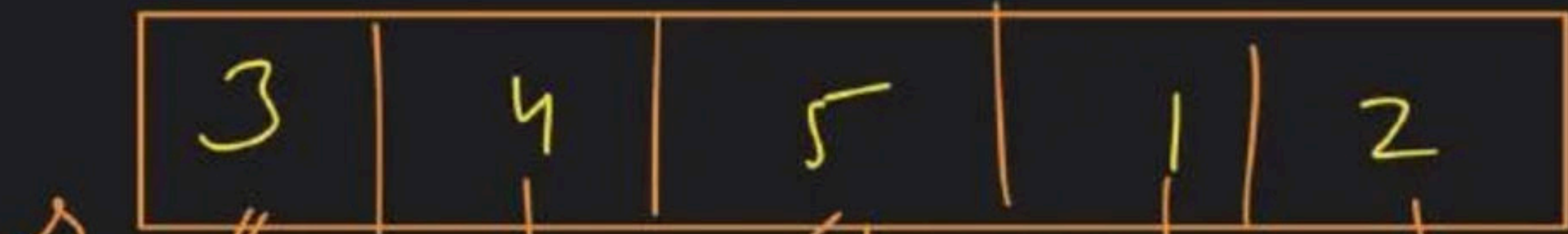
+ 1
bal

Total deficit = 6 lit

✓



km/
out



$gas < km$
↓
deficit = 2

$gas < km$
↓
deficit = 2

$gas < km$
↓
deficit = $|3 - 1| = 2$

$gas > km$
↓
bal = $|4 - 1| = 3$

$gas > km$
↓
bal = $|5 - 2| = 3$

~~deficit = 6~~
~~balance = 6~~
~~start = 3~~

deficit = 6, balance = 6, start = 3

if (bal - deficit $\geq$ 0)

↳ circle complete $\rightarrow$ return start

else

↳ circle can't be completed $\rightarrow$ return -1